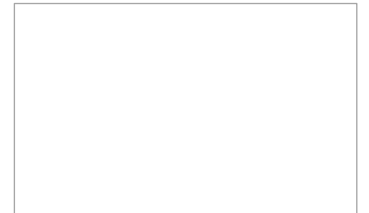




QpiAI™ - Joint AI Training and Certification Program

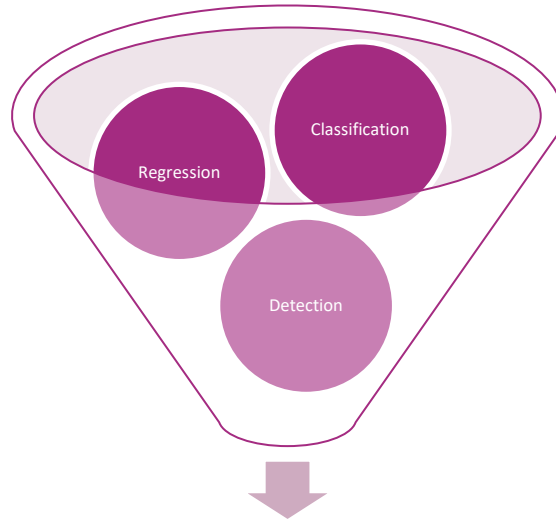
Introduction to Machine Learning, Linear Regression, Logistic Regression and K Nearest Neighbors

Sachin Kumar



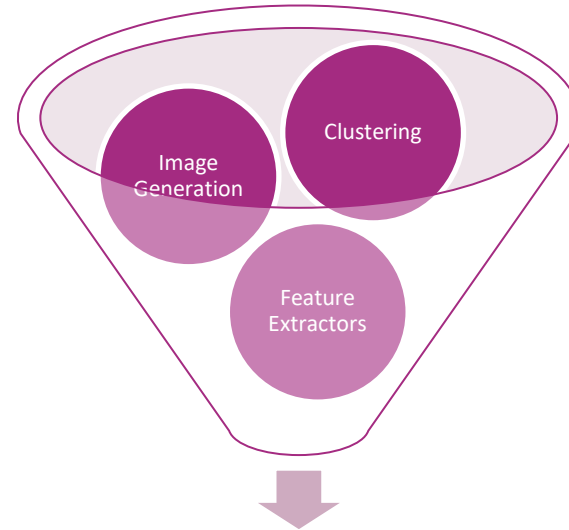
Sachin Kumar

Introduction to Machine Learning



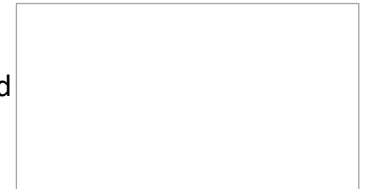
Supervised

For each observation of the predictor measurement(s) x_i , $i = 1, \dots, n$ there is an associated response measurement y_i

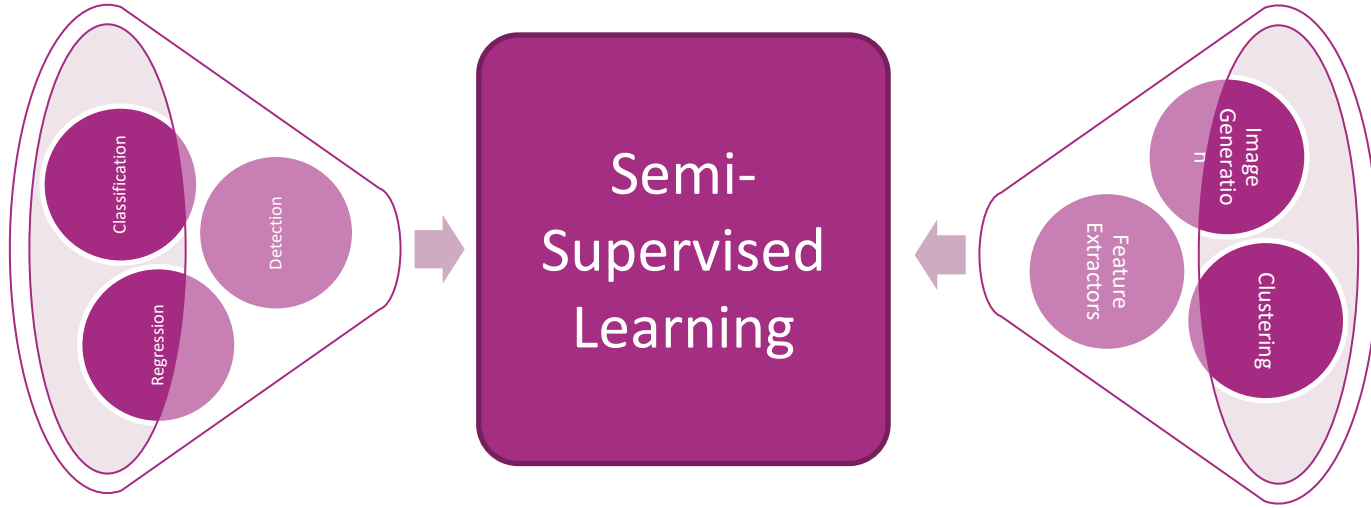


Unsupervised

For every observation $i = 1, \dots, n$, we observe a vector of measurements x_i but no associated response y_i

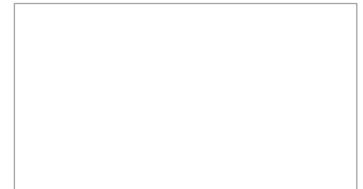


Introduction to Machine Learning

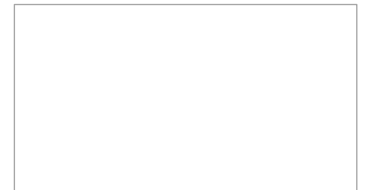
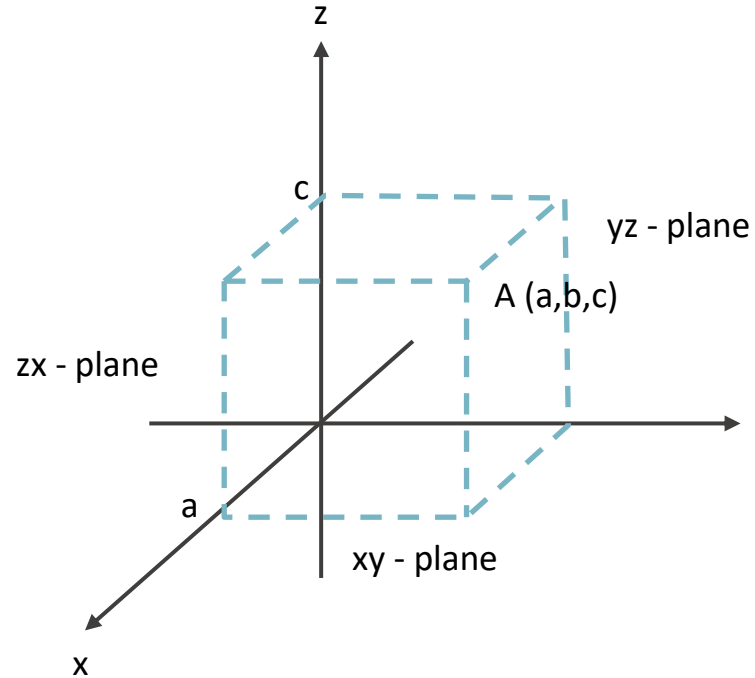


For building a universal handwriting recognizer

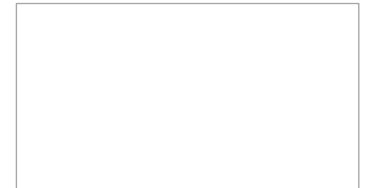
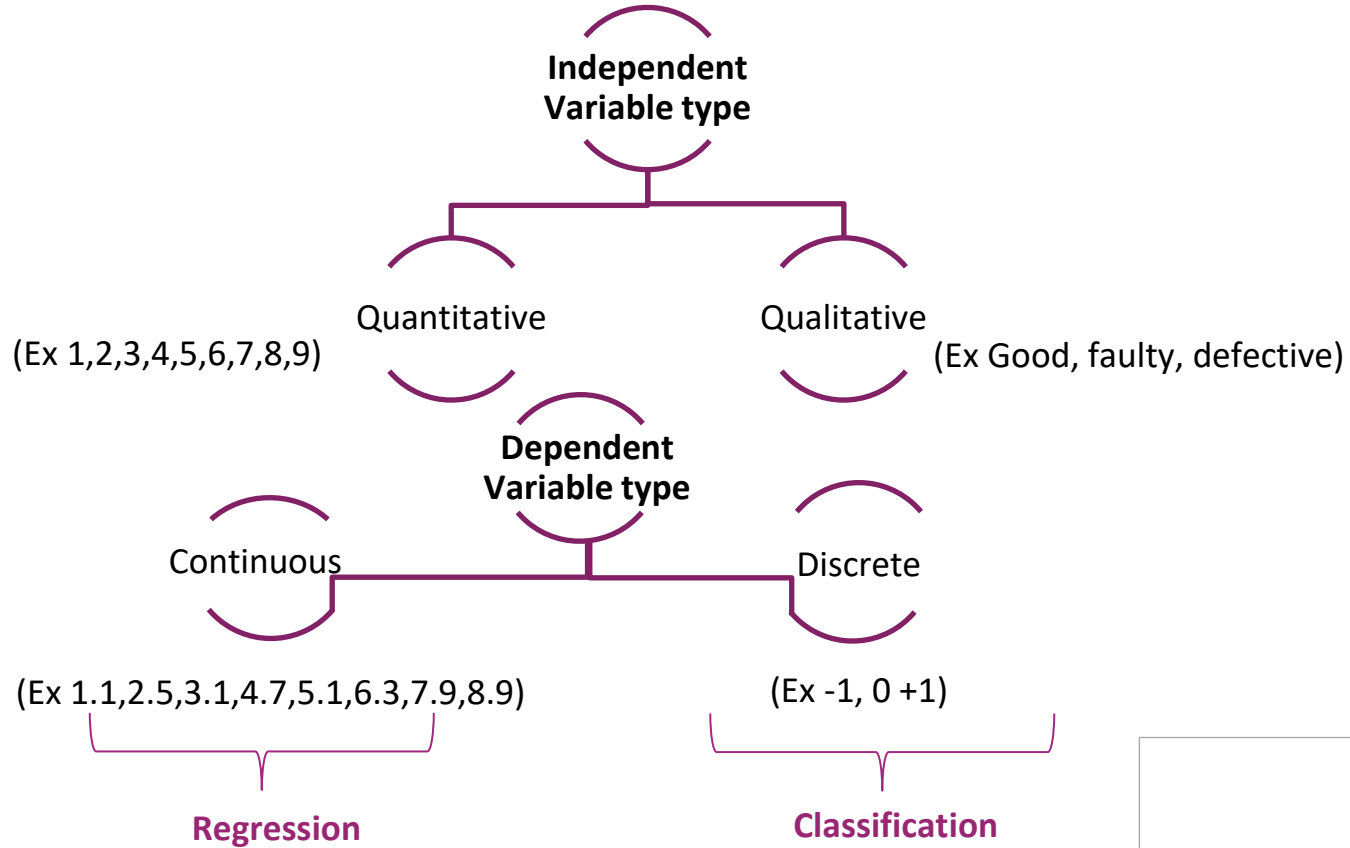
- Train a weak Classifier
- Generate Data using existing labels
- Now use all the data to train a robust classifier



Independent vs Dependent Variable



What is Regression and Classification?



Nominal

A variable with categories that does not have a natural order or ranking. Can code nominal variables with numbers, but the order is arbitrary and any calculations, such as computing a mean, median, or standard deviation, is meaningless. Eg: blood group, zip code, ethnicity, eye color.

Ordinal

An ordinal scale is one where the order matters but not the difference between values. Eg:

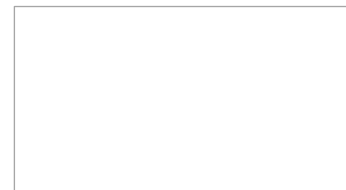
- Socio economic status: low income, middle income , high income
- Education level: high school, undergraduate, graduate, doctoral

Interval

A variable following interval scale is one where there is order and the difference between two values is meaningful. Eg: Temperature in Celsius, pH, Credit Score.

Ratio

A ratio variable, has all the properties of an interval variable, and also has a clear definition of 0. When the variable equals 0, there is none of that variable.
Eg: Weight, Temperature in Kelvin.



Regression and Classification Models

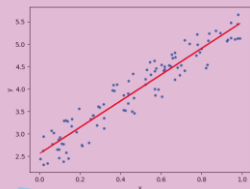
*** Predicting Hemoglobin in blood using PPG signal**

1. Peaks And Troughs
2. Heart beat rate
3. Power spectrum
4. Age
5. Gender

Regression

Linear
Regression

Multiple Linear
Regression



Classification

Logistic
regression

K-Nearest
Neighbour

*** Patients Heart disease prediction**
tobacco consumption,
cholesterol,
alcohol consumption,
type a personality traits,
adiposity value,
obesity measures,
systolic blood pressure

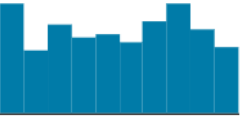
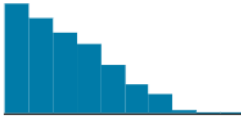
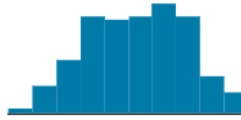
*** Weather forecasting applications for predicting rainfall and weather conditions**
1. Temperature last day
2. Rainfall recorded
3. Sunlight recorded hrs
4. Humidity recorded



*** Predicting Engine remaining life**
1. Multiphysics sensor data
2. Make and made of sensor
3. Material of engine
4. Hours of engine usage

*** In voting applications to find out whether voters will vote for a particular candidate or not**
1. Number of promises fulfilled
2. Popularity in voters
3. Familiarity with regional problems

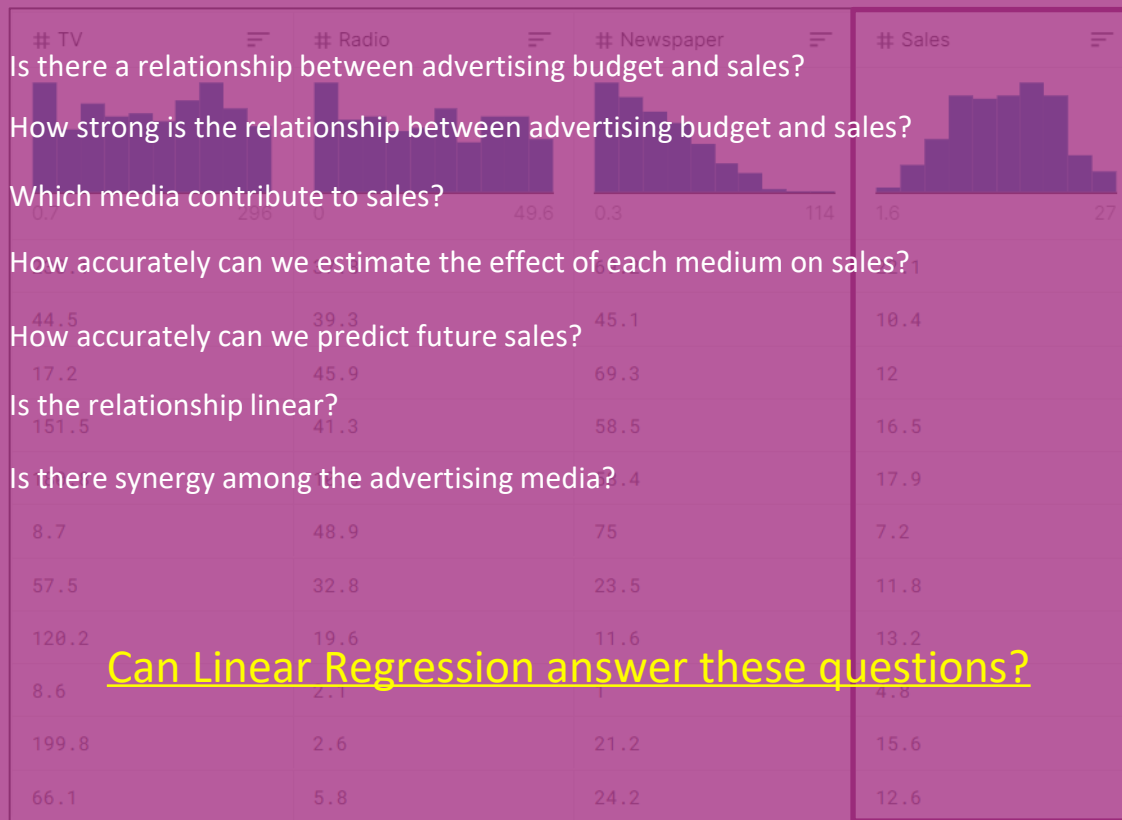
Motivation for Linear Regression

# TV 	# Radio 	# Newspaper 	# Sales 
 0.7 296	 0 49.6	 0.3 114	 1.6 27
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	12
151.5	41.3	58.5	16.5
180.8	10.8	58.4	17.9
8.7	48.9	75	7.2
57.5	32.8	23.5	11.8
120.2	19.6	11.6	13.2
8.6	2.1	1	4.8
199.8	2.6	21.2	15.6
66.1	5.8	24.2	12.6

<https://www.kaggle.com/ashydv/advertising-dataset>

Motivation for Linear Regression

1. Is there a relationship between advertising budget and sales?
2. How strong is the relationship between advertising budget and sales?
3. Which media contribute to sales?
4. How accurately can we estimate the effect of each medium on sales?
5. How accurately can we predict future sales?
6. Is the relationship linear?
7. Is there synergy among the advertising media?



Can Linear Regression answer these questions?

<https://www.kaggle.com/ashydv/advertising-dataset>

Simple Linear Regression using a single predictor X

- We assume a model:

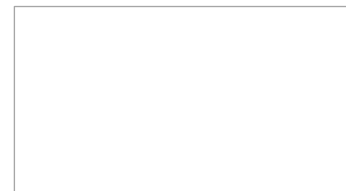
$$Y = \beta_0 + \beta_1 X + \epsilon$$

where β_0 and β_1 are two unknown constants that represent the **intercept** and **slope**, also known as **coefficients** or **parameters**, and ϵ is the error term.

- Given some estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ for the model **coefficients**, we predict future sales using

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x,$$

where $\hat{y}$ indicates a prediction of Y on the basis of $X = x$.
The **hat** symbol denotes an estimated value.



Estimation of the parameters by Least Squares

- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be the prediction for Y based on the i th value of X. Then $e_i = y_i - \hat{y}_i$ represents the i^{th} **residual**
- We define the **residual sum of squares (RSS)** as

$$\text{RSS} = e_1^2 + e_2^2 + \cdots + e_n^2,$$

or equivalently as

$$\text{RSS} = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \cdots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2$$

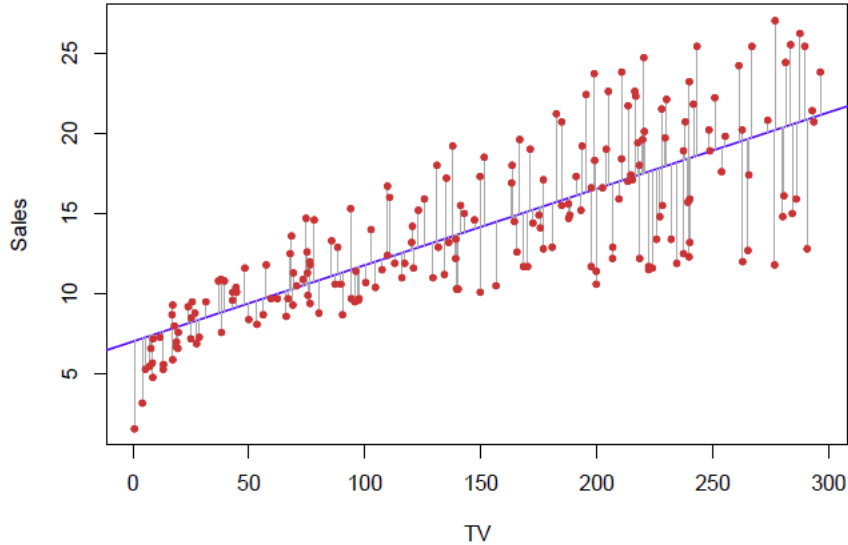
- The least squares approach chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the RSS. The minimizing values can be shown to be

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

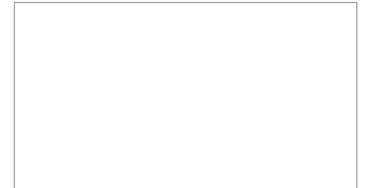
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

Where $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ are the sample means.

Example: Advertising data

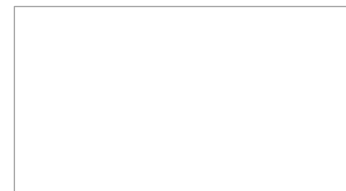


The least squares fit for the regression of **sales** onto **TV**. In this case a linear fit captures the essence of the relationship, although it is somewhat deficient in the left of the plot.



Assumptions of Linear Regression

- **Linear Relationship between the features and target :** Linear regression captures only linear relationships between the features and target. It can be validated by plotting Scatter Plot between features and the target.
- **Little or no Multicollinearity between the features :** Multicollinearity is a state of very high inter-correlations or inter-associations among the independent variables. It is therefore a type of disturbance in the data if present weakens the statistical power of the regression model. Pair plots and Heatmaps (Correlation Matrix) can be used for identifying highly correlated features.
- **Homoscedasticity Assumption :** Homoscedasticity describes a situation in which the error term (that is, the “noise” or random disturbance in the relationship between the features and the target) is the same across all values of the independent variables. A scatter plot of residual values v/s predicted values is a good way to check for homoscedasticity.
- **Normal distribution of error terms :** The fourth assumption is that the error(residuals) follow a normal distribution. Normal distribution of the residuals can be validated by plotting a q-q plot. The residuals should not have high correlations with the predictors.
- **Little or No autocorrelation in the residuals :** Autocorrelation occurs when the residual errors are dependent on each other. The presence of correlation in error terms drastically reduces model's accuracy. This usually occurs in time series models where the next instant is dependent on previous instant. Autocorrelation can be tested with the help of Durbin-Watson test.



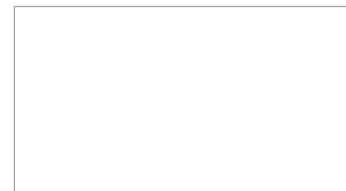
Multiple Linear Regression

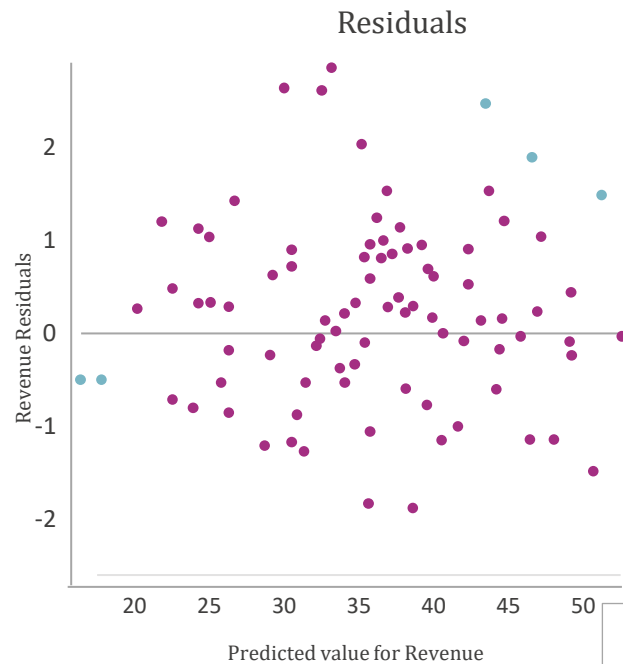
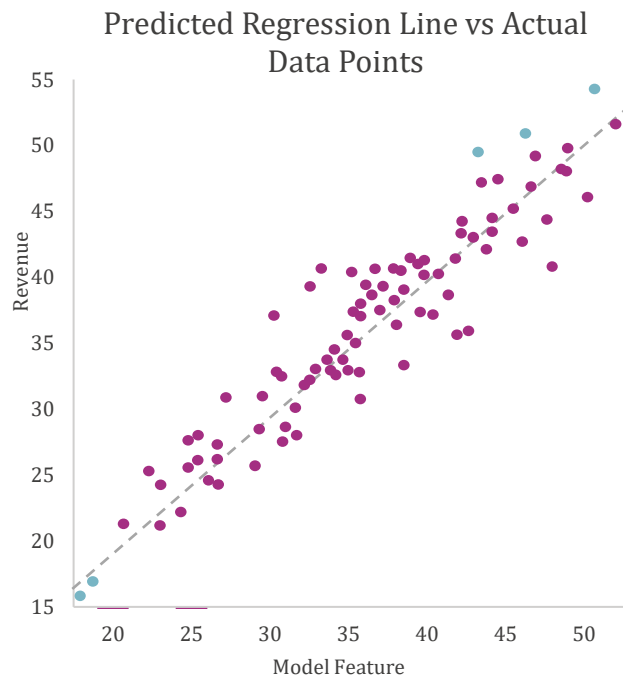
- Here our model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon,$$

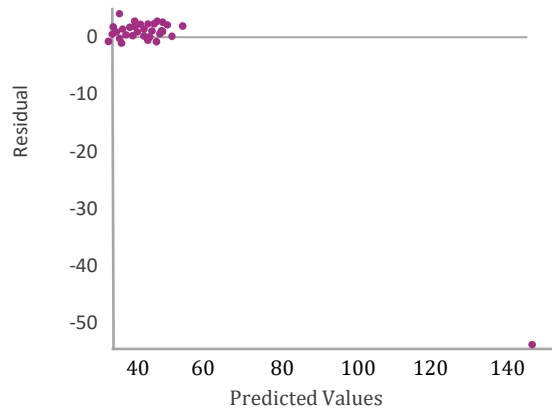
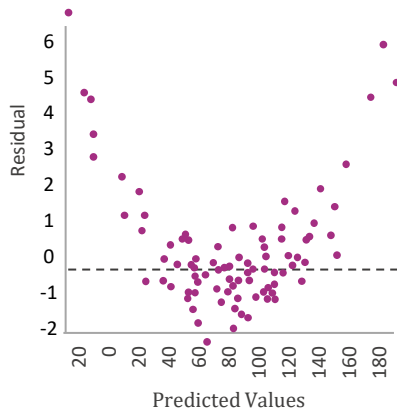
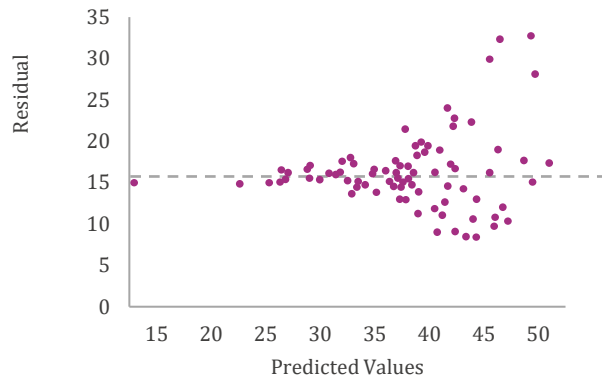
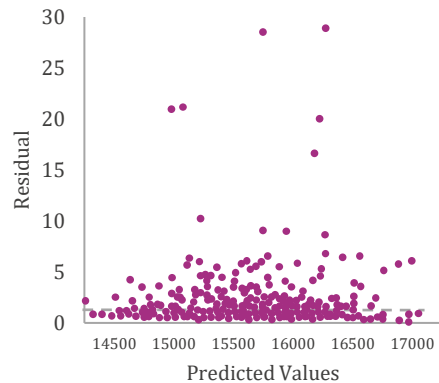
- We interpret β_j as the **average** effect on Y of a one unit increase in X_j , **holding all other predictors fixed**.
In the advertising example, the model becomes

$$\text{Sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \epsilon$$



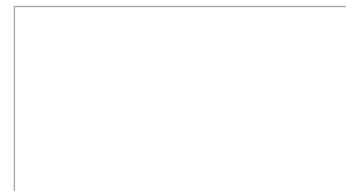
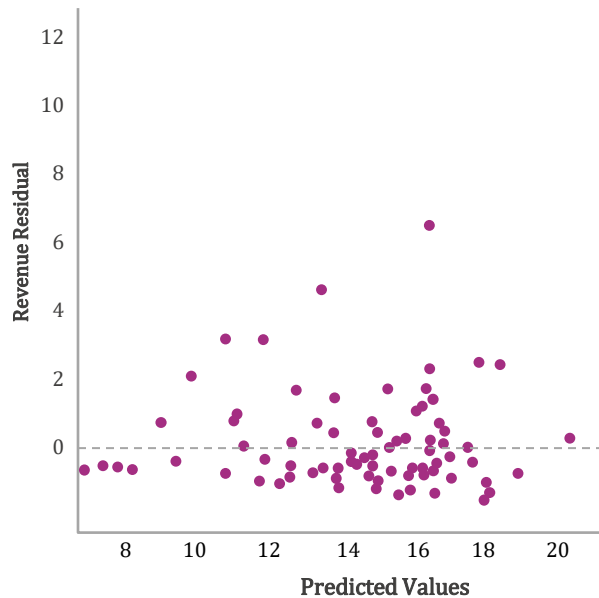
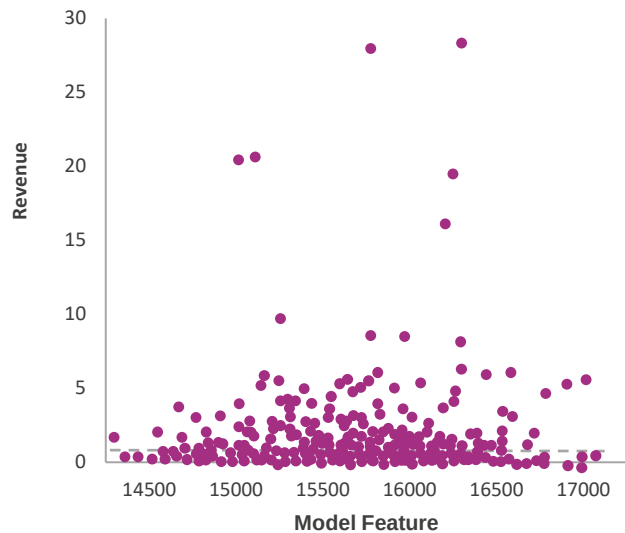


Non-Ideal Residuals

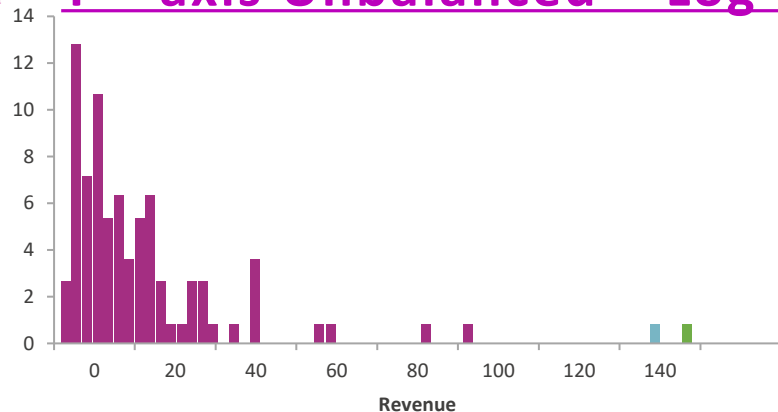


Y – axis Unbalanced

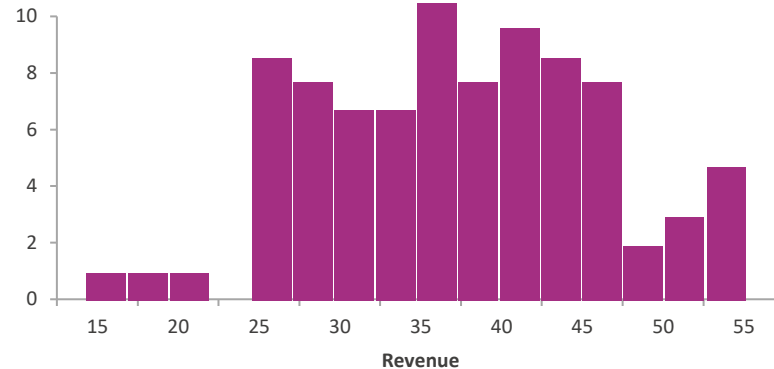
Predicted Regression Line vs Actual
Data Points



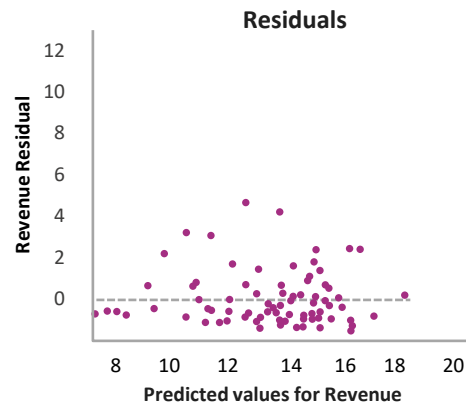
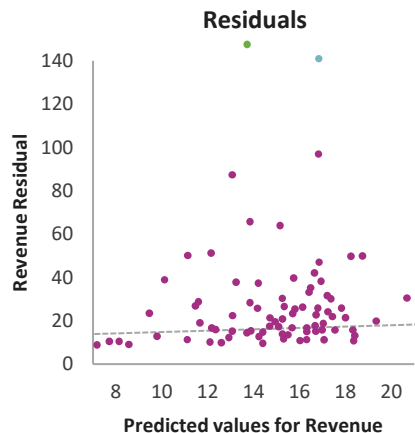
Y – axis Unbalanced – Log Tranform of Y

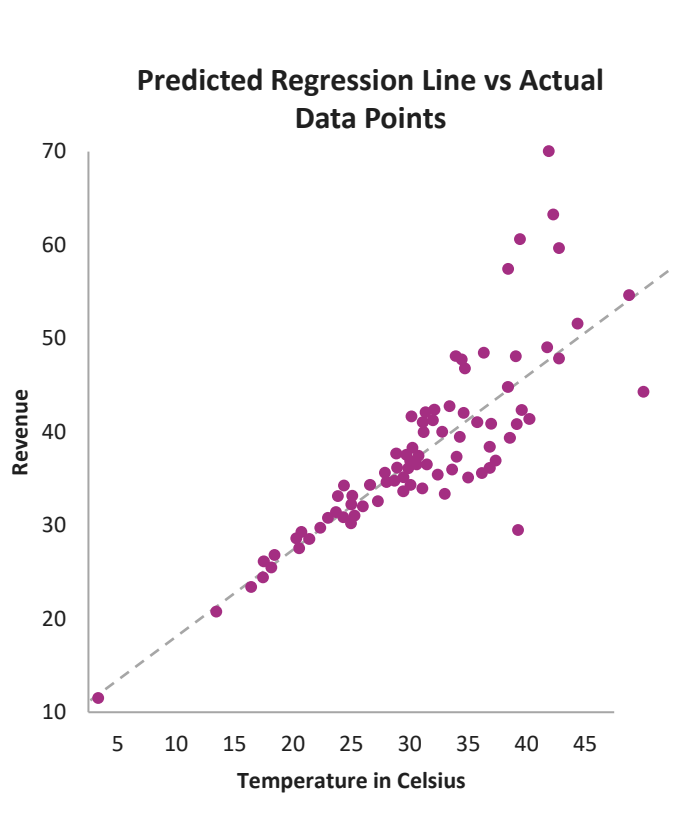


Not very close to Normal distribution.

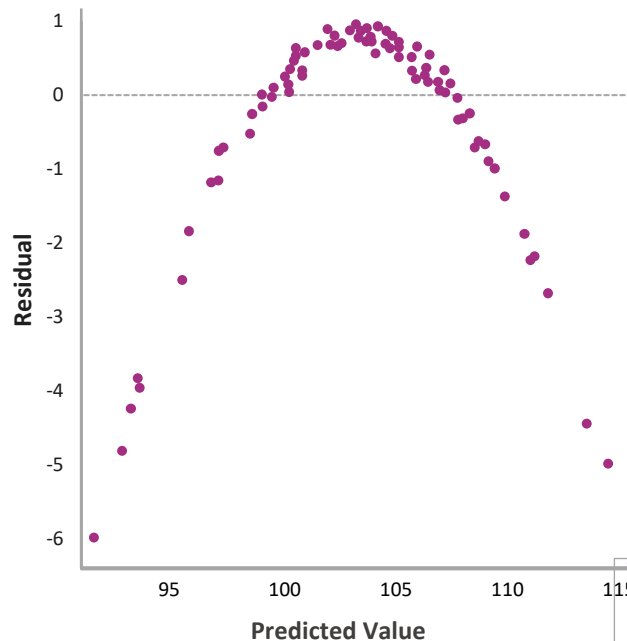
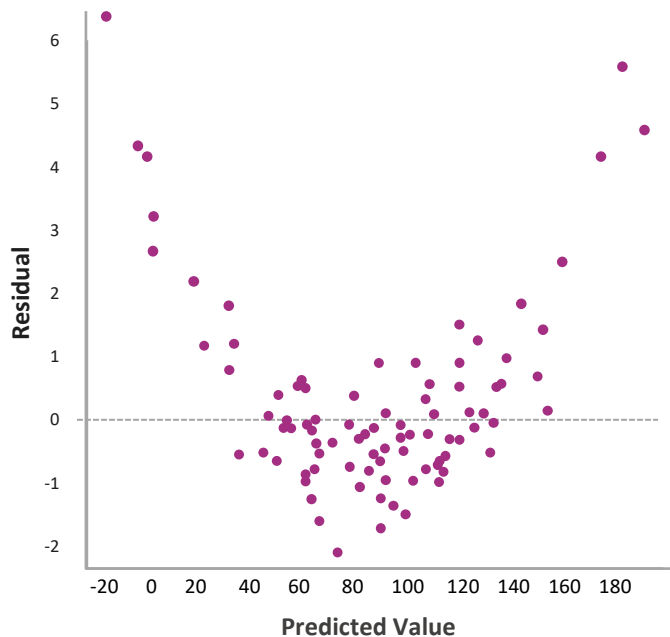


Closer to Normal distribution by transformation.

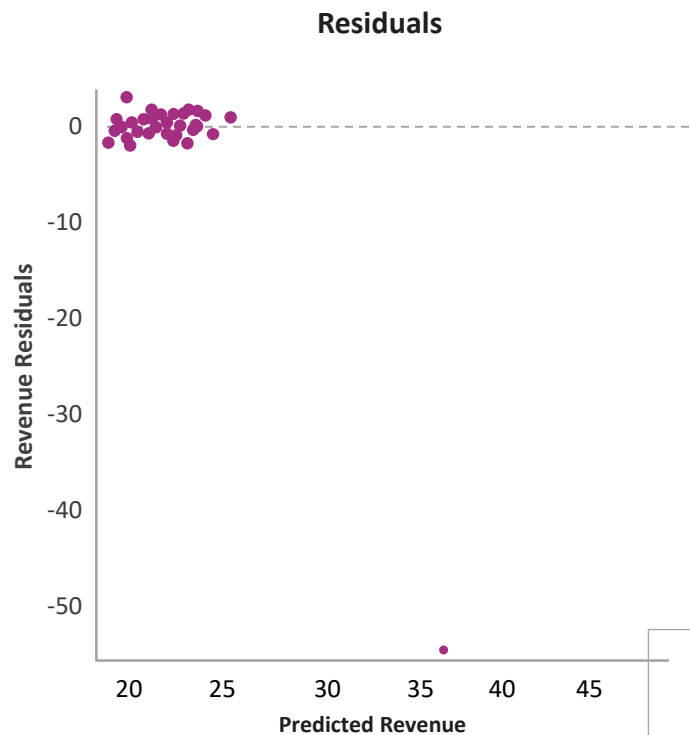
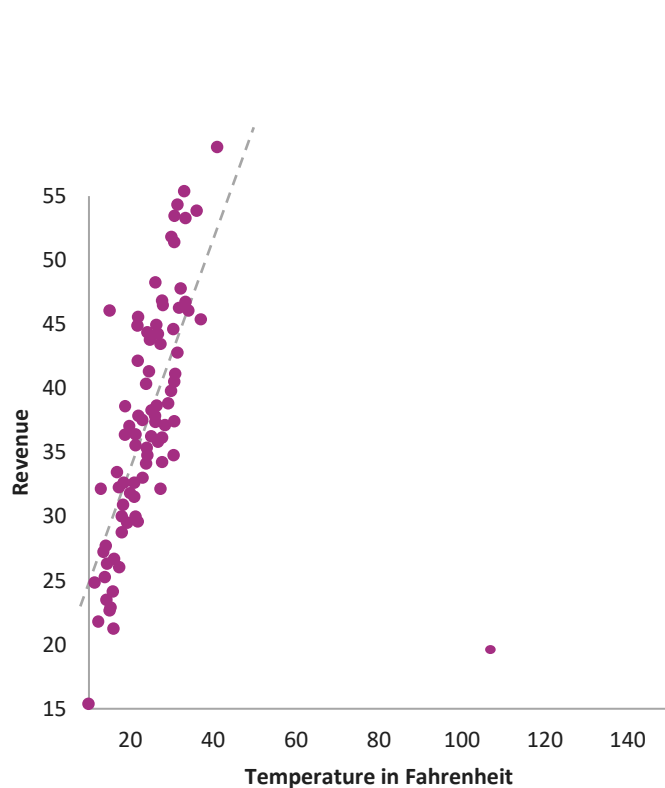




Non-Linear Relationship – Higher Order Polynomial

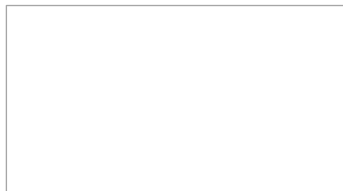


Outlier – Remove Outlier Datapoints



Important pre-conditions for Linear Regression

- **Residuals should be identically distributed:** Each residual is sampled from the same normal distribution $N(0, \sigma^2)$, with a mean of zero and the same variance throughout.
- **Any predictor variable must not be correlated with the residuals (Independent):** If a feature is related to the residuals, then that feature can predict the residuals. The feature's weight would need to be adjusted/ variable added to the model.
- **Neighboring residuals must not be correlated (Independent):** If adjacent residuals are correlated, one residuals can predict the next residual, as in autocorrelation. Use the Durbin-Watson test to assess autocorrelation.
- **Residuals must have a constant variance:** Heteroscedasticity is the case where residuals have a non-constant variance.
- **Key Transforms:** For a feature x : x^2 , x^3 , $\log(x)$, $\log(ax + b)$, $1/(x + a)$, $1/\text{square_root}(x)$, Box-Cox transforms etc.



Standard errors can also be used to perform hypothesis tests on the coefficients. The most common hypothesis test involves testing the null hypothesis of:

H_0 : There is no relationship between X and Y versus the alternative hypothesis

H_A : There is some relationship between X and Y

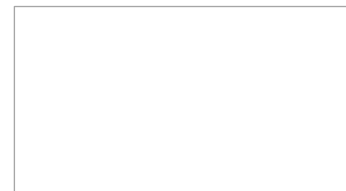
Mathematically, this corresponds to testing:

- $H_0 : \beta_1 = 0$
- $H_A : \beta_1$ not equal to 0

Since if: $\beta_1 = 0$, then the model reduces to $Y = \beta_0 + \epsilon$; thus X is not associated with Y .

To test the null hypothesis, we compute a t-statistic, and then compute the p-value, where:

- $$\text{t-statistic} = (\hat{\beta}_1 - 0) / \text{SE}(\hat{\beta}_1)$$



Assessing the Accuracy of the Coefficient Estimates

- The standard error of an estimator reflects how it varies under repeated sampling. We have

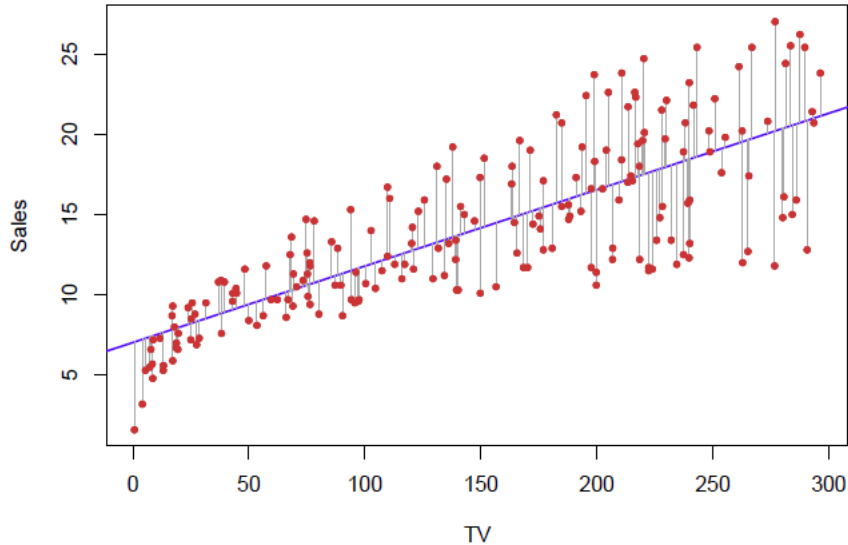
$$\text{SE } (\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \text{SE } (\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

Where $\sigma^2 = \text{Var}(\epsilon)$

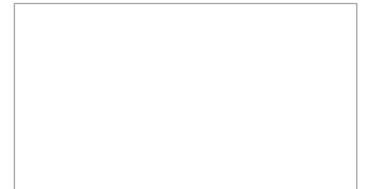
- These standard errors can be used to compute **coefficient intervals**. A 95% confidence interval is defined as a range of values such that with 95% probability, the range will contain the true unknown value of the parameter. It has the form

$$\hat{\beta}_1 \pm 2 \cdot \text{SE}(\hat{\beta}_1).$$

Example: Advertising data



The least squares fit for the regression of **sales** onto **TV**. In this case a linear fit captures the essence of the relationship, although it is somewhat deficient in the left of the plot.

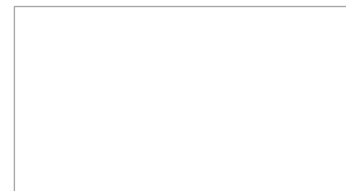


That is, there is approximately a 95% chance that the interval

$$[\hat{\beta}_1 - 2 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot SE(\hat{\beta}_1)]$$

will contain the true value of β_1 (under a scenario where we got repeated samples like the present sample)

For the advertising data, the 95% confidence interval for β_1 is [0.042, 0.053]



Estimation and Prediction for Multiple Regression

- Given estimates $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$, we can make predictions using the formula

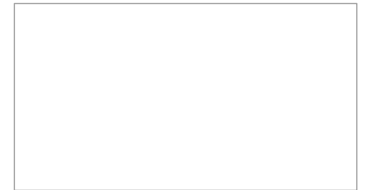
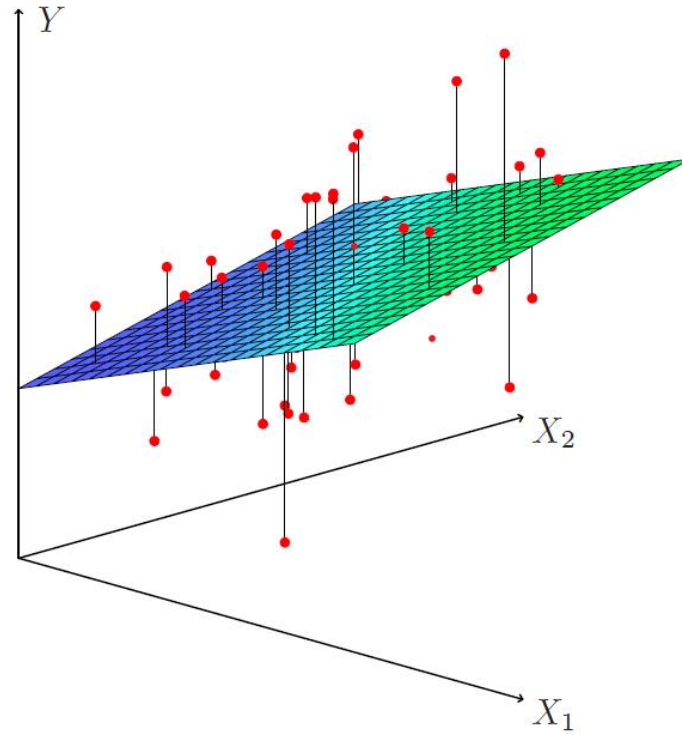
$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_p x_p$$

- Given estimates $\beta_0, \beta_1, \dots, \beta_p$ as the values that minimize the sum of squared residuals

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \dots - \hat{\beta}_p x_{ip})^2$$

The values $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ that minimize RSS are the multiple least squares regression coefficient estimates.



Assessing the Overall Accuracy of the Model

- We compute the **Residual Standard Error**:

$$RSE = \sqrt{\frac{1}{n-2} RSS} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y})^2},$$

where the **residual sum-of-squares** is
 $RSS = \sum_{i=1}^n (y_i - \hat{y})^2$

- R-squared** or fraction of variance explained is:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

Where $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$ is the **total sum of squares**.

- Adjusted R-squared** or fraction of variance explained after factoring in multiple predictors:

$$Adjusted R^2 = 1 - \frac{\frac{RSS}{n-p-1}}{\frac{TSS}{n-1}} = 1 - \frac{(1 - R^2)(n-1)}{n-p-1}$$

Where p is the number of **Predictors**.

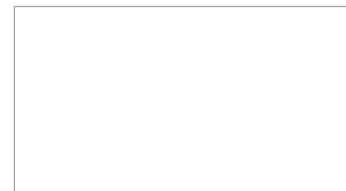
Multiple Linear Regression

- Here our model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon,$$

- We interpret β_j as the **average** effect on Y of a one unit increase in X_j , **holding all other predictors fixed**.
In the advertising example, the model becomes:

$$\text{Sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \epsilon$$



Results for Advertising Data

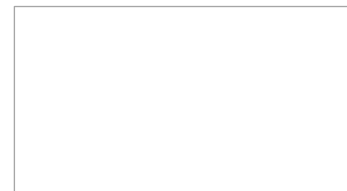
	Coefficient	Std. Error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
Radio	0.189	0.0086	21.89	< 0.0001
Newspaper	-0.001	0.0059	-0.18	0.8599

Correlation Matrix

	TV	Radio	Newspaper	Sales
TV	1.0000	0.0548	0.0567	0.7822
Radio		1.0000	0.3541	0.5762
Newspaper			1.0000	0.2283
Sales				1.0000

Interpreting regression coefficients

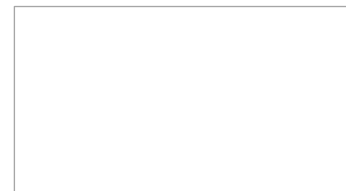
- The ideal scenario is when the predictors are uncorrelated - **a balanced design**:
 - Each coefficient can be estimated and tested separately.
 - Interpretations such as “**a unit change in X_j is associated with a β_j change in Y , while all the other variables stay fixed**”, are possible.
- Correlations amongst predictors cause problems:
 - The variance of all coefficients tends to increase, sometimes dramatically
 - Interpretations become hazardous - when X_j changes, everything else changes.
- **Claims of causality** should be avoided for observational data.



The woes of (interpreting) regression coefficients

“Data Analysis and Regression” Mosteller and Tukey 1977

- A regression coefficient β_j estimates the expected change in Y per unit change in X_j , **with all other predictors held fixed**. But predictors usually change together!
- Let Y = number of tackles by a football player in a season
- W and H are his weight and height. Fitted regression model is $\hat{Y} = b_0 + 0.50W - 0.10H$
- How do we interpret $\bar{\beta}_2 < 0$?



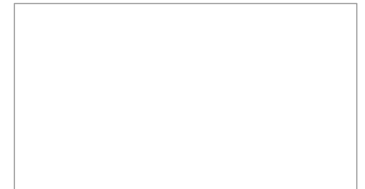
Qualitative predictors with more than two levels

- Let us consider a case of investigating the difference in Credit Card balance of individuals with different ethnicity. For qualitative predictors with more than two levels, we create additional dummy variables. For example, for the **ethnicity** variable we create two dummy variables. The first could be:

$$x_{i1} = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ person is Asian} \\ 0 & \text{if } i^{\text{th}} \text{ person is not Asian} \end{cases}$$

and the second could be:

$$x_{i2} = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ person is Caucasian} \\ 0 & \text{if } i^{\text{th}} \text{ person is not Caucasian} \end{cases}$$

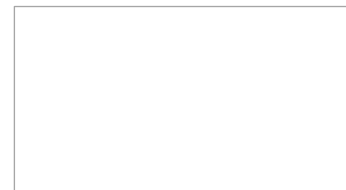


Qualitative predictors with more than two levels - continued

- Then both of these variables can be used in the regression equation, in order to obtain the model

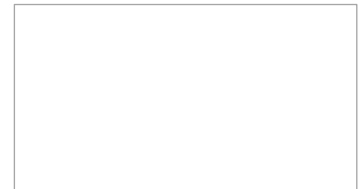
$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if } i^{\text{th}} \text{ person is Asian} \\ \beta_0 + \beta_2 + \epsilon_i & \text{if } i^{\text{th}} \text{ person is Caucasian} \\ \beta_0 + \epsilon_i & \text{if } i^{\text{th}} \text{ person is African American} \end{cases}$$

- There will always be one fewer dummy variable than the number of levels. The level with no dummy variable - African American in this example - is known as the **baseline**.



Results for ethnicity

	Coefficient	Std. Error	t-statistic	p-value
Intercept	531.00	46.32	11.464	< 0.0001
ethnicity[Asian]	-18.69	65.02	-0.287	0.7740
ethnicity[Caucasian]	-12.50	56.68	-0.221	0.8260



Extensions of the Linear Model

Interactions:

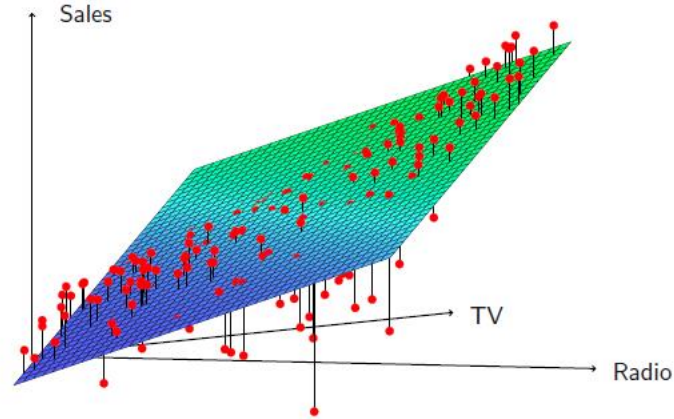
- In our analysis of the **Advertising** data, we assumed that the effect on **sales** of increasing one advertising medium is independent of the amount spent on the other media.
- For example, the linear model:

$$\hat{y} = \beta_0 + \beta_1 \times \mathbf{TV} + \beta_2 \times \mathbf{radio} + \beta_3 \times \mathbf{newspaper}$$

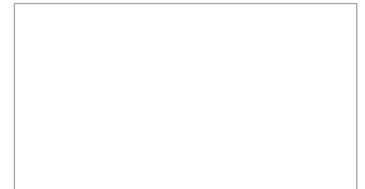
states that the average effect on **sales** of a one-unit increase in **TV** is always β_1 , regardless of the amount spent on **radio**.

- But suppose that spending money on radio advertising actually increases the effectiveness of TV advertising, so that the slope term for **TV** should increase as **radio** increases.
- In this situation, given a fixed budget of \$100,000, spending half on **radio** and half on **TV** may increase **sales** more than allocating the entire amount to either **TV** or to **radio**.
- In marketing, this is known as a **synergy** effect, and in statistics it is referred to as **interaction** effect.

Interaction in the Advertising data?



- When levels of either **TV** or **radio** are low, then the true **Sales** are lower than predicted by the linear model. But when advertising is split between the two media, then the model tends to underestimate **Sales**.



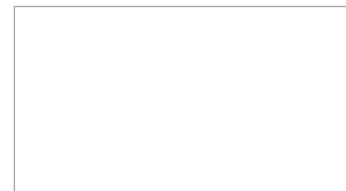
Model takes the form

$$\begin{aligned}\text{Sales} &= \beta_0 + \beta_1 \times \mathbf{TV} + \beta_2 \times \mathbf{radio} + \beta_3 \times (\mathbf{radio} \times \mathbf{TV}) + \epsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \times \mathbf{radio}) \times \mathbf{TV} + \beta_2 \times \mathbf{radio} + \epsilon\end{aligned}$$

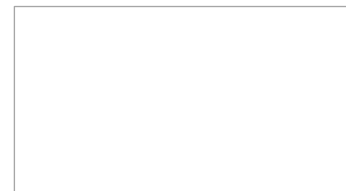
Results:

	Coefficient	Std. Error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

- The results in this table suggests that interactions are important.
- The p-value for the interaction term **TV** $\times$ **radio** is extremely low, indicating that there is strong evidence for $H_A: \beta_3 \neq 0$.
- The R^2 for the interaction model is 96.8%, compared to only 89.7% for the model that predicts **sales** using **TV** and **radio** without an interaction term.



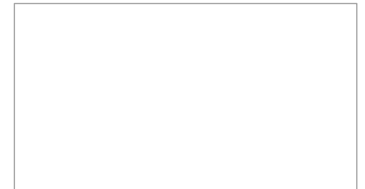
- This means that $(96.8 - 89.7)/(100 - 89.7) = 69\%$ of the variability in **sales** that remains after fitting the additive model has been explained by the interaction term.
- The coefficient estimates in the table suggest that an increase in TV advertising of \$1,000 is associated with increased sales of
 $(\hat{\beta}_1 + \hat{\beta}_3 \times \mathbf{radio}) \times 1000 = 19 + 1.1 \times \mathbf{radio}$ units.
- An increase in radio advertising of \$1,000 will be associated with an increase in sales of
 $(\hat{\beta}_2 + \hat{\beta}_3 \times \mathbf{TV}) \times 1000 = 29 + 1.1 \times \mathbf{TV}$ units.



- Sometimes it is the case that an interaction term has a very small p-value, but the associated main effects (in this case, **TV** and **radio**) do not.

The hierarchy principle:

- If we include an interaction in a model, we should also include the main effects, even if the p-values associated with their coefficients are not significant.
- The rationale for this principle is that interactions are hard to interpret in a model without main effects – their meaning is changed.
- Specifically, the interaction terms also contain main effects, if the model has no main effect terms.



Interactions between qualitative and quantitative variables

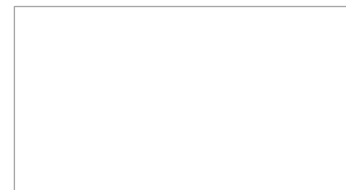
- Consider the **Credit** details of individuals, and suppose that we wish to predict **balance** using **income** (quantitative) and whether an individual is **student** (qualitative).
- Without an interaction term, the model takes the form

$$\begin{aligned} \text{balance}_i &\approx \beta_0 + \beta_1 \times \text{income}_i + \begin{cases} \beta_2 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases} \\ &= \beta_1 \times \text{income}_i + \begin{cases} \beta_0 + \beta_2 & \text{if } i\text{th person is a student} \\ \beta_0 & \text{if } i\text{th person is not a student} \end{cases} \end{aligned}$$

Interactions between qualitative and quantitative variables

With interactions, it takes the form

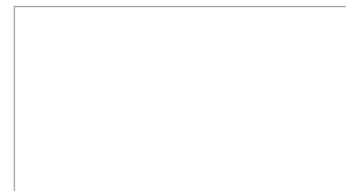
$$\begin{aligned} \text{balance}_i &\approx \beta_0 + \beta_1 \times \text{income}_i + \begin{cases} \beta_2 + \beta_3 \times \text{income}_i & \text{if student} \\ 0 & \text{if not student} \end{cases} \\ &= \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \times \text{income}_i & \text{if student} \\ \beta_0 + \beta_1 \times \text{income}_i & \text{if not student} \end{cases} \end{aligned}$$



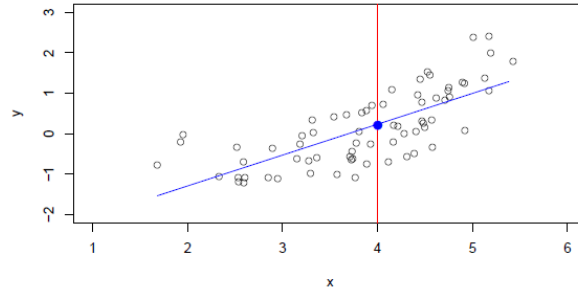
The **linear** model is an important example of a parametric model:

$$f_L(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

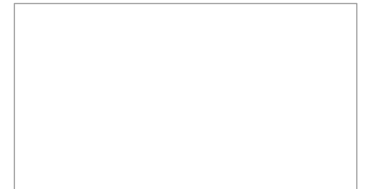
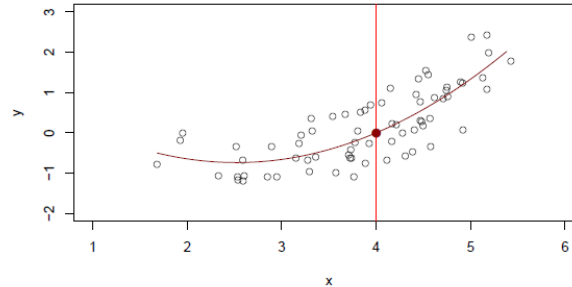
- A linear model is specified in terms of $p + 1$ parameters $\beta_0 + \beta_1, \dots, \beta_p$.
- We estimate the parameters by fitting the model to training data.
- Although it is **almost never correct**, a linear model often serves as a good and interpretable approximation to the unknown true function $f(X)$.

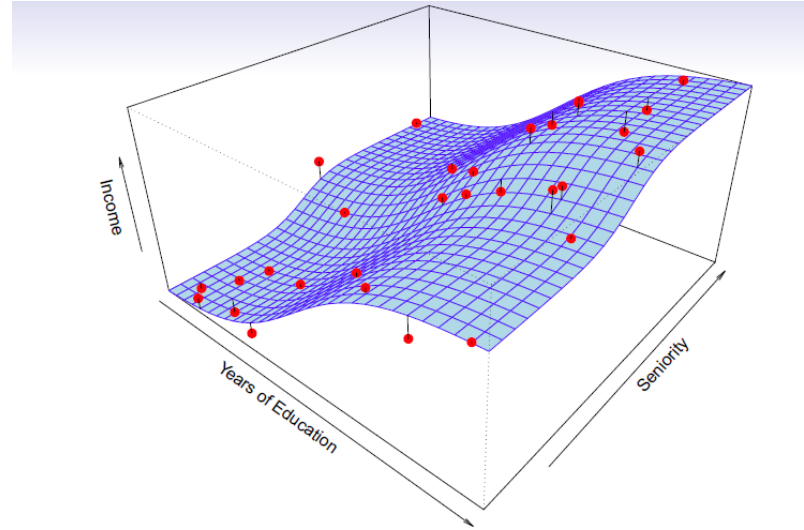


A linear model $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ gives a reasonable fit here



A quadratic model $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$ fits slightly better.

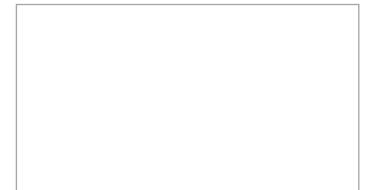


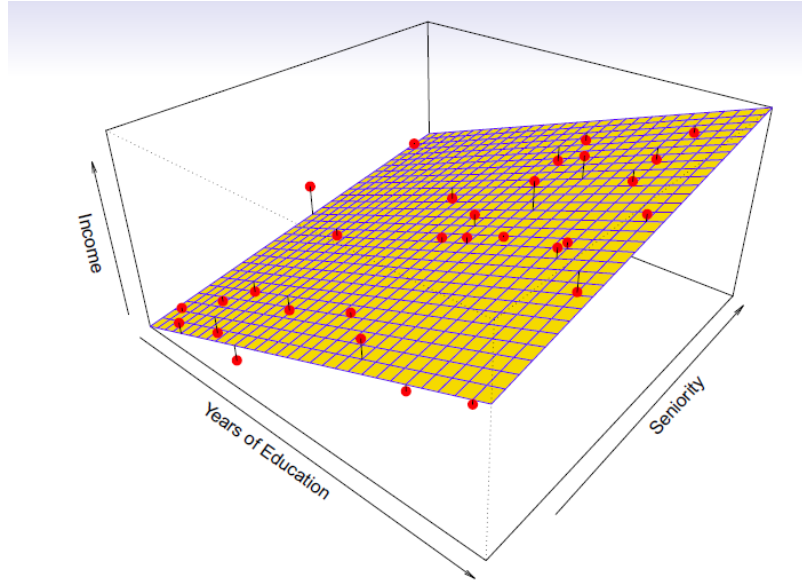


Simulated example. Red points are simulated values for **income** from the model

$$\mathbf{income} = f(\mathbf{education}, \mathbf{seniority}) + \epsilon$$

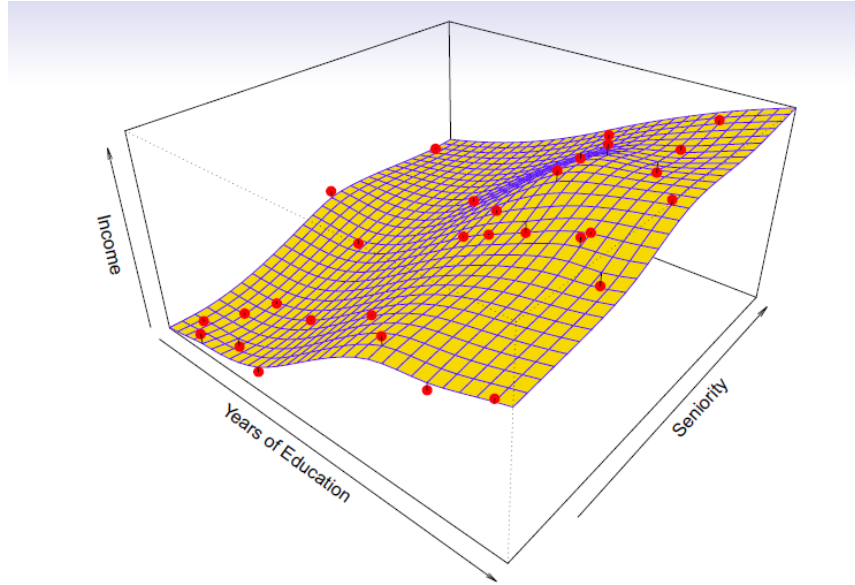
f is the blue surface.



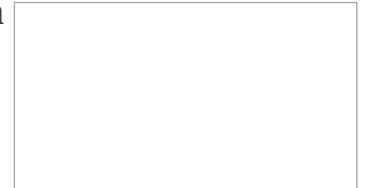


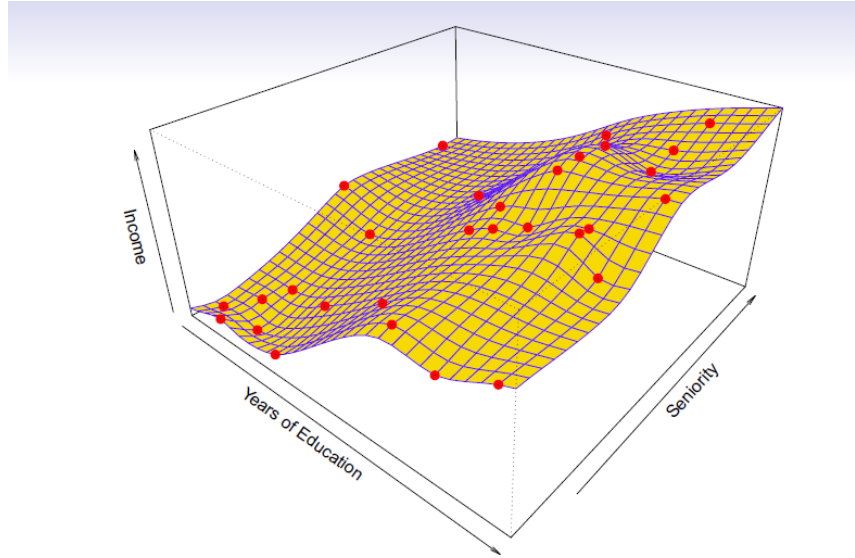
Linear regression model fit to the simulated data.

$$\hat{f}_L(\text{education, seniority}) = \hat{\beta}_0 + \hat{\beta}_1 \times \text{education} + \hat{\beta}_2 \times \text{seniority}$$

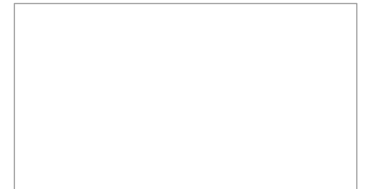


More flexible regression model $\hat{f}_s(\text{education}, \text{seniority})$ fit to the simulated data. Here we use a technique called a **thin-plate spline** to fit a flexible surface. We control the roughness of the fit





Even more flexible spline regression model $\hat{f}_s(\text{education}, \text{seniority})$ fit to the simulated data. Here the fitted model makes no errors on the training data! Also known as **overfitting**.



Train Test Split (Random and Stratified Sampling)

Now we've understood how the Linear Regression model works, how do we know that it will work properly any outside data other than the training dataset?

To know it better we don't train on model on the entire dataset, instead we take a split of it. On one the split we train our model (train set) and the other split is emulated as the outside world data on which we evaluate the model (test set).

Usually on a large dataset we usually split such a way that 80% of datapoints are in train set and the rest 20% is test set.

Some classification problems do not have a balanced number of examples for each class label. As such, it is desirable to split the dataset into train and test sets in a way that preserves the same proportions of examples in each class as observed in the original dataset. This is when we use Stratified Sampling.



Suppose we fit a model $\hat{f}(x)$ to some training data

$\text{Tr} = \{x_i, y_i\}_1^N$, and we wish to see how well it performs.

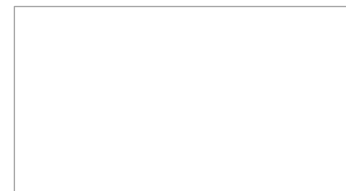
- We could compute the average squared prediction error over Tr:

$$\text{MSE}_{\text{Tr}} = \text{Ave}_{i \in \text{Tr}} [y_i - \hat{f}(x_i)]^2$$

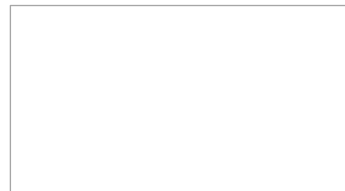
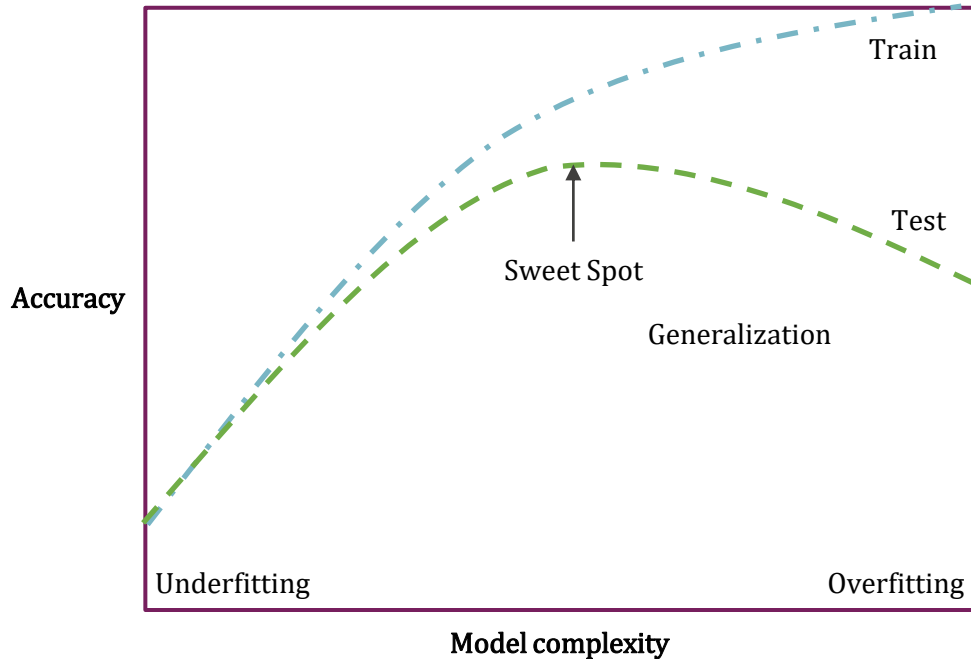
This could be biased toward more overfit models.

- Instead we should, compute it using fresh **test** data $\text{Te} = \{x_i, y_i\}_1^M$:

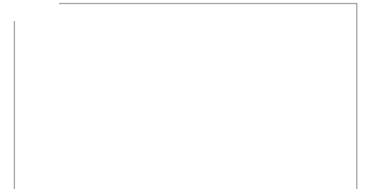
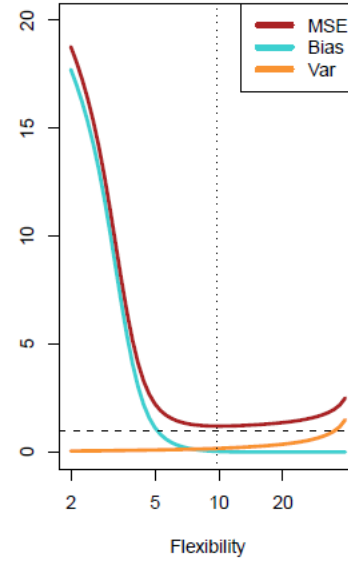
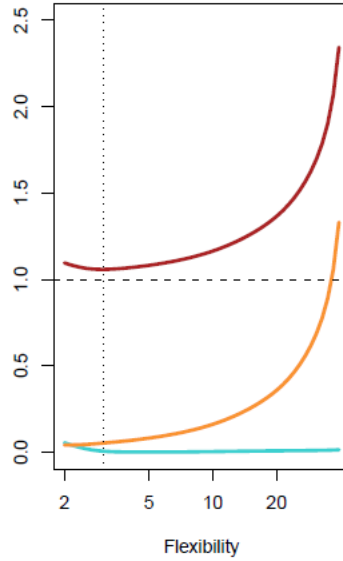
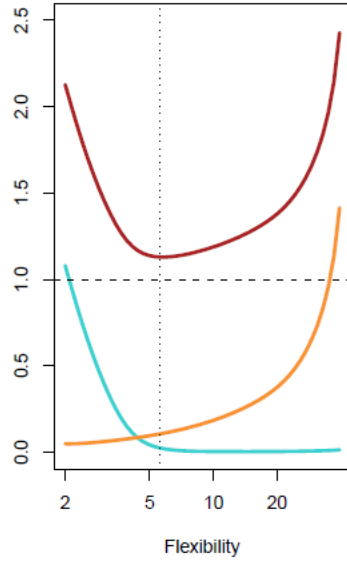
$$\text{MSE}_{\text{Te}} = \text{Ave}_{i \in \text{Te}} [y_i - \hat{f}(x_i)]^2$$

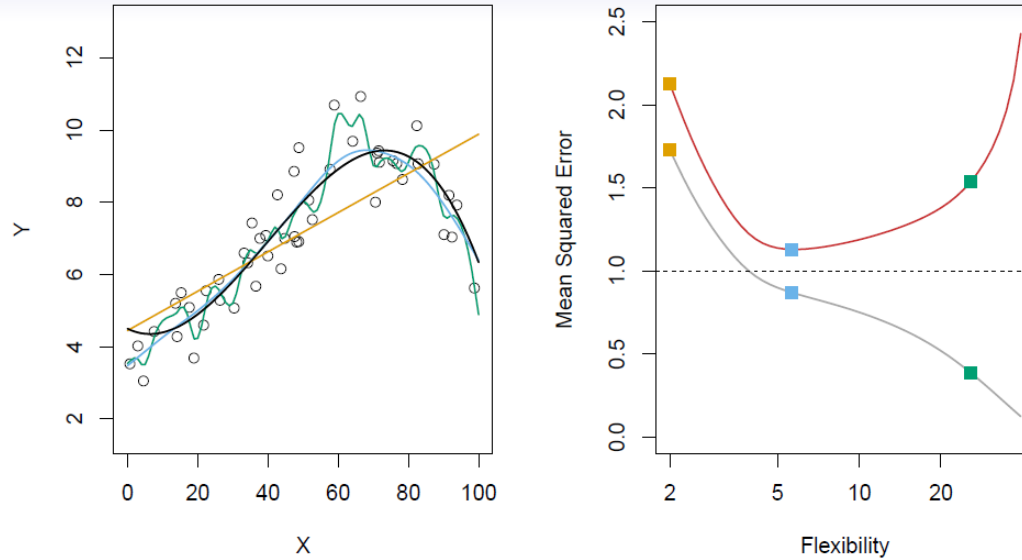


Identifying Model Performance Sweet Spot

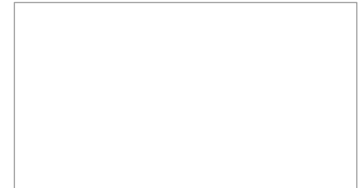


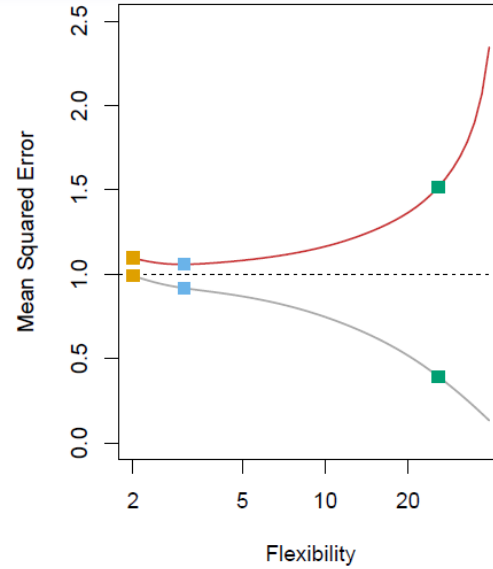
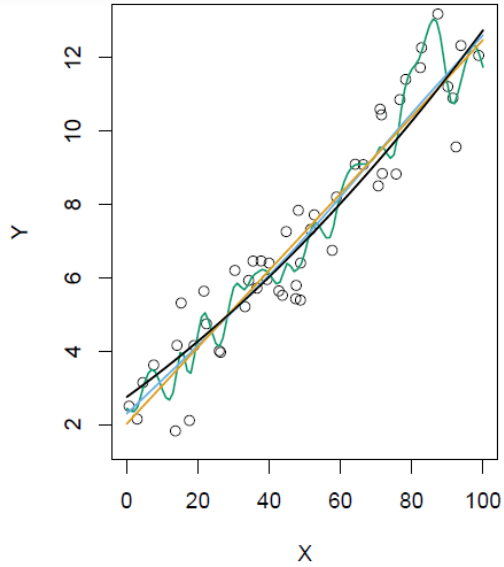
Bias-variance trade-off for 3 examples



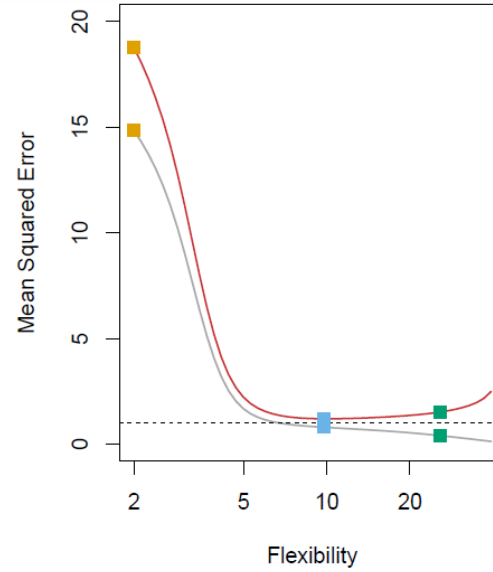
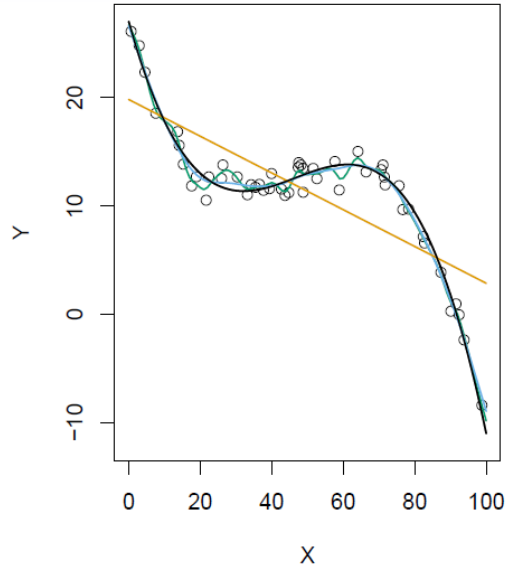


Black curve is truth. Red curve on right is MSE_{Te} , grey curve is MSE_{Tr} . Orange, blue and green curves/squares correspond to fits of different flexibility.



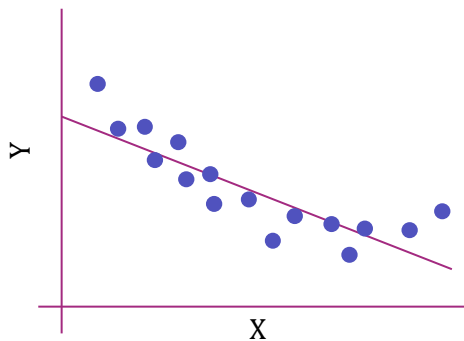


Here the truth is smoother, so the smoother fit and linear model do really well.



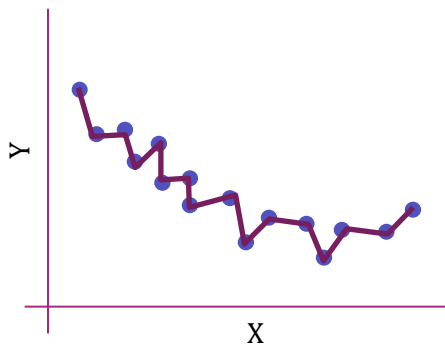
Here the truth is wiggly and the noise is low, so the more flexible fits do the best.

Underfit



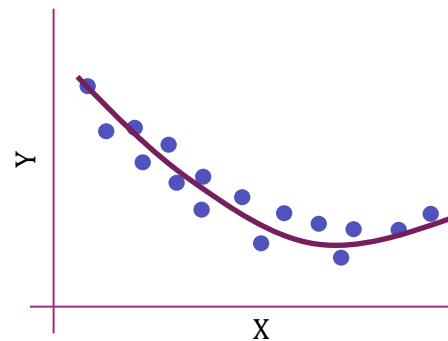
$$Y = \beta_0 + \beta_1 * X$$

Overfit

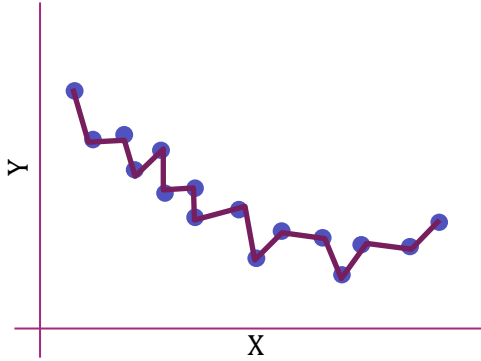


$$Y = \beta_0 + \beta_1 * X + \beta_2 * X^2 + \beta_3 * X^3 + \beta_4 * X^4 + \dots$$

Balanced fit



$$Y = \beta_0 + \beta_1 * X + \beta_2 * X^2$$



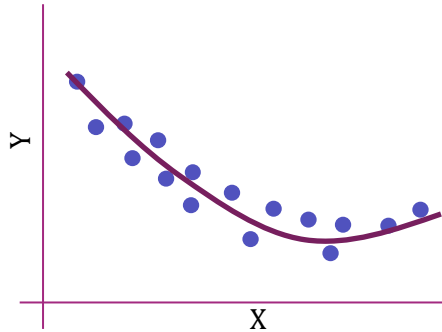
$$Y = \beta_0 + \beta_1 * X + \beta_2 * X^2 + \beta_3 * X^3 + \beta_4 * X^4 + \dots$$



Try to make β_3 , β_4 and other parameters almost close to zero



$$Y = \beta_0 + \beta_1 * X + \beta_2 * X^2$$



L2 Regularization (Ridge Regression)

Suppose we penalize and make $\beta_1, \beta_2, \beta_3, \beta_4$ small

$$\text{Original Loss} = \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$\min_{\beta} \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 + 1000 \beta_1^2 + 1000 \beta_2^2 + 1000 \beta_3^2 + 1000 \beta_4^2$$

L2 Regularization Formula

$$J(\beta) = \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^n \beta_j^2$$

- Whenever we try to increase the value of β_i , the error term increases so we add a penalty term which forces the β_i to approach towards zero.
- Lambda is the regularization hyper-parameter whose value is optimized for best results

L1 Regularization (Lasso Regression)

Suppose we penalize and make $\beta_1, \beta_2, \beta_3, \beta_4$ small and close to zero

$$\min_{\beta} \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$\min_{\beta} \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 + 1000 |\beta_1| + 1000 |\beta_2| + 1000 |\beta_3| + 1000 |\beta_4|$$

L2 Regularization Formula

$$J(\beta) = \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^n |\beta_j|$$

- In L1 Regularization, we penalize the absolute value of the weights.
- Unlike L2 Regularization, the weights may be reduced to zero here.
- Hence, it is very useful when we are trying to compress our model/performing feature selection.

Standard Scaling

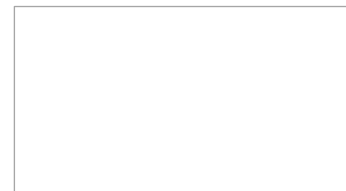
Standardization is another scaling technique where the values are centered around the mean with a unit standard deviation by computing Z-score. This means that the mean of the attribute becomes zero and the resultant distribution has a unit standard deviation.

$$x_s = \frac{x - \mu}{\sigma}$$

Feature Normalization or Min-Max Scaling

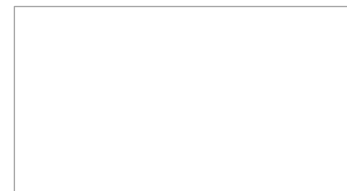
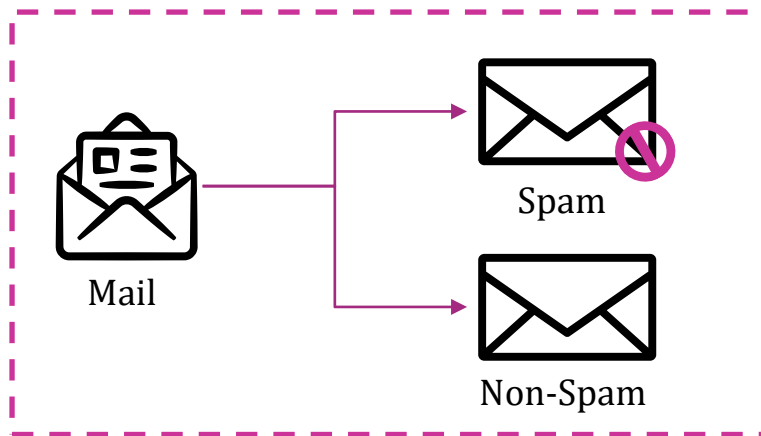
Normalization is a scaling technique in which values are shifted and rescaled so that they end up ranging between 0 and 1. It is also known as Min-Max scaling.

$$x_N = \frac{x - x_{min}}{x_{max} - x_{min}}$$



What is Classification ?

- Classification is a process of categorizing a given set of data into classes.
- The process starts with predicting the class of given data points. The classes are often referred to as target, label or categories.
- The classification predictive modeling is the task of approximating the mapping function from input variables to discrete output variables. The main goal is to identify which class/category the new data will fall into.



Real World Use Cases Of Classification

How can we classify covid-19 from other respiratory diseases?

- Fever
- Loss of smell
- Shortness of breath
- Chest Pain
- Dry Cough

How does Google classify whether a user will be interested a particular content?

- Ads on which users clicked recently
- Frequently visited sites

How do medical researchers classify whether a tumor is malignant or not?

- Nature of tumour
- Affected body area
- Family History

- Logistic Regression is used for classification problems i.e where the response is qualitative or categorical. For example: binary classification problems.
- The 'sigmoid' function that is used for logistic regression is as follows:

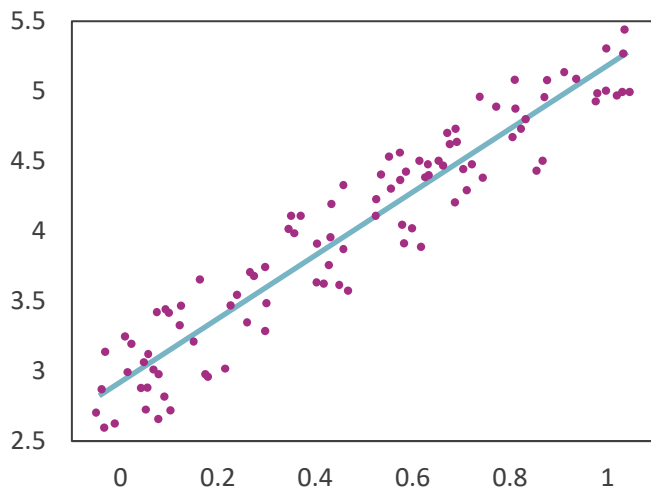
$$p(x) = \frac{e^{\theta_0 + \theta_1 x}}{1 + e^{\theta_0 + \theta_1 x}}$$

$$p(x) = \frac{e^{h_\theta(x)}}{e^{h_\theta(x)} + 1} = \frac{1}{1 + e^{-h_\theta(x)}}$$

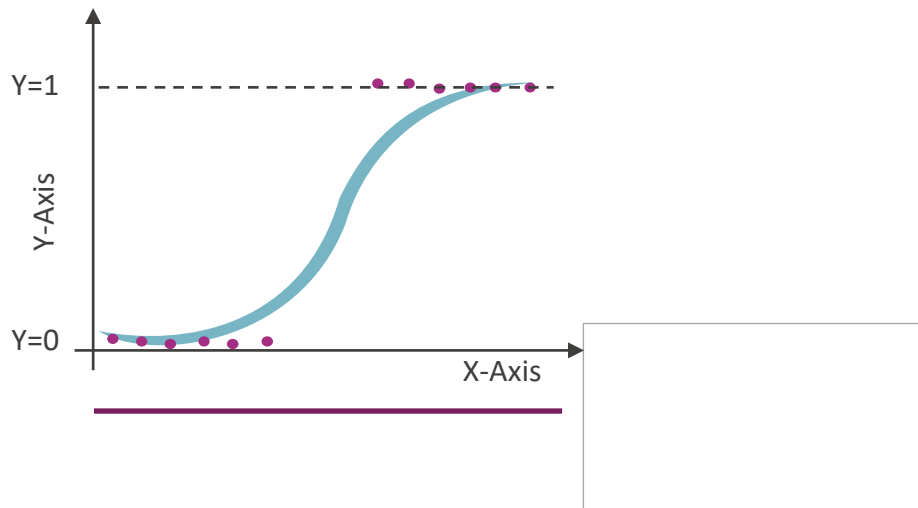
Q π AI Why Not Linear Regression?

- Linear regression is the type of regression we use for a continuous, normally distributed response variable.
- Logistic regression is the type of regression we use for a binary response variable that follows a Bernoulli distribution

Linear Regression



Logistic Regression



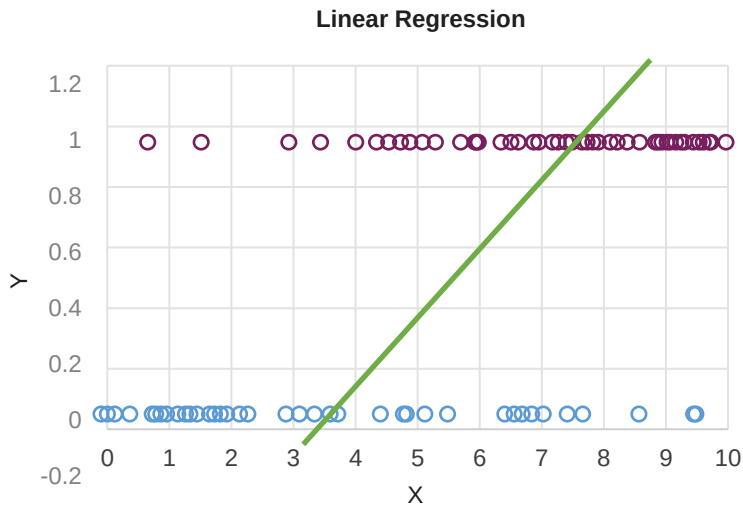
Modeling binary outcomes with Linear Regression

Fitting a linear regression model on a binary outcome Y :

- $Y_i | X_i \sim \text{Bernoulli}(p(x_i))$
- $E(Y_i) = \beta_0 + \beta_1 X_i = \hat{p}(x_i)$

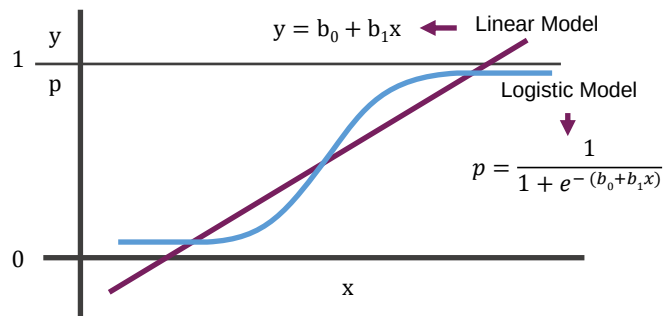
Problems?

- Linear relationship between X and Y ?
- Normally distributed errors?
- Constant variance of Y ?
- Is $\hat{p}$ guaranteed to be in $[0,1]$?



Why can't we use linear regression for binary outcomes?

- The relationship between X and Y is not linear.
- The response Y is not normally distributed.
- The variance of a Bernoulli random variable depends on its expected value p.
- Fitted value of Y may not be 0 or 1, since linear models produce fitted values in $(-\infty, \infty)$



Measure	Min	Max	Name
$P(Y = 1)$	0	1	"Probability"
$\frac{P(Y = 1)}{1 - P(Y = 1)}$	0	∞	"Odds"
$\text{Log} \left[\frac{P(Y = 1)}{1 - P(Y = 1)} \right]$	$-\infty$	∞	"Log-Odds" or "logit"

- The odds of an event are defined as:

$$\text{odds}(Y = 1) = \frac{P(Y = 1)}{P(Y = 0)} = \frac{P(Y = 1)}{1 - P(Y = 1)} = \frac{p}{1 - p}$$

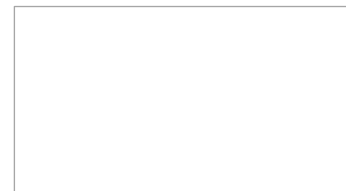
$$p = \frac{\text{odds}(Y = 1)}{1 + \text{odds}(Y = 1)}.$$

Understanding Odds Ratio

- Odds Ratios (OR) can be useful for comparisons.
- Suppose we have a trial to see if an intervention T reduces mortality, compared to a placebo, in patients with high cholesterol. The odds ratio is

$$OR = \frac{\text{odds (death|intervention T)}}{\text{odds(death|placebo)}}$$

- The OR describes the benefits of intervention T:
 - $OR < 1$: the intervention is better than the placebo since $\text{odds(death|intervention T)} < \text{odds(death|placebo)}$
 - $OR = 1$: there is no difference between the intervention and the placebo.
 - $OR > 1$: the intervention is worse than the placebo since $\text{odds(death|intervention T)} > \text{odds(death|placebo)}$



Logistic Regression Model

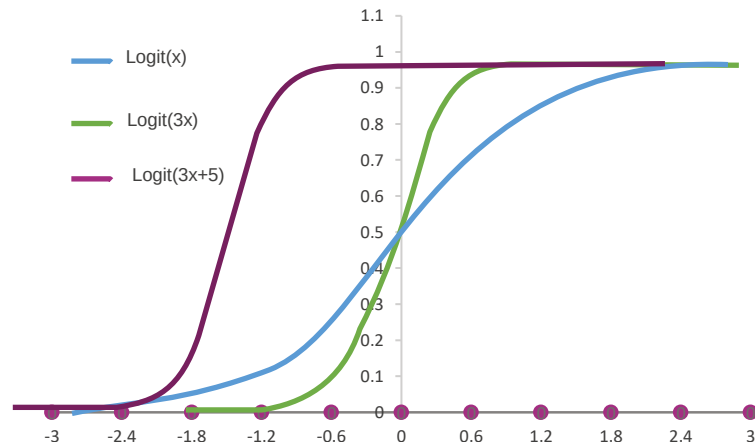
- Let Y be a binary outcome and X a predictor.
- We are interested in modelling $p_x = P(Y=1|X=x)$, i.e. the probability of success for the predictor value of $X=x$.

Define the **Logistic Regression model** as:

$$\text{logit}(p(x)) = \log\left(\frac{p(x)}{1-p(x)}\right) = \theta_0 + \theta_1 X$$

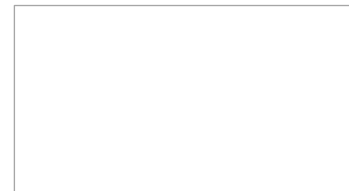
- $\log\left(\frac{p(x)}{1-p(x)}\right)$ is called the **logit** function
- $p(x) = \frac{e^{\theta_0 + \theta_1 X}}{1 + e^{\theta_0 + \theta_1 X}}$
- $\lim_{x \rightarrow -\infty} \frac{e^x}{1+e^x} = 0$ and $\lim_{x \rightarrow \infty} \frac{e^x}{1+e^x} = 1$, so $0 \leq p(x) \leq 1$.

Scale & location shift: logit ($Ax+B$)



$\frac{p(x)}{1-p(x)}$ is called the **odds** of success,

$\log\left(\frac{p(x)}{1-p(x)}\right)$ is called the **log odds** of success.



- Logistic Regression is used for classification problems i.e where the response is qualitative or categorical. For example: binary classification problems.
- The 'sigmoid' function that is used for logistic regression is as follows:

$$p(x) = \frac{e^{\theta_0 + \theta_1 x}}{1 + e^{\theta_0 + \theta_1 x}}$$

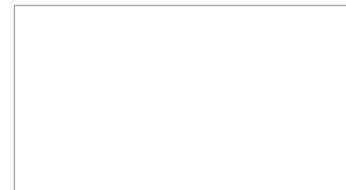
$$p(x) = \frac{e^{h_\theta(x)}}{e^{h_\theta(x)} + 1} = \frac{1}{1 + e^{-h_\theta(x)}}$$

Interpretation Odds Ratios and predicted probability in Logistic Regression

- $OR > 1$: positive relationship: as X increases, the probability of Y increases; greater X is associated with higher odds of positive outcome.
- $OR < 1$: negative relationship: as X increases, probability of Y decreases; greater X is associated with lower odds of positive outcome.
- $OR = 1$: no association with X
- For a data sample, if the predicted probability is 0.7, then it means the model predicts that the sample belongs to **class 1 with probability 0.7** and to **class 0 with probability 0.3**.
- Similarly, if the predicted probability for a sample is 0.2, then it means the model predicts that the sample belongs to **class 1 with probability 0.2** and to **class 0 with probability 0.8**.

In logistic regression, we test null hypothesis of the form:

- $H_0 : \beta_1 = 0$ (which corresponds to $OR = 1$)
- $H_A : \beta_1$ is not equal to 0



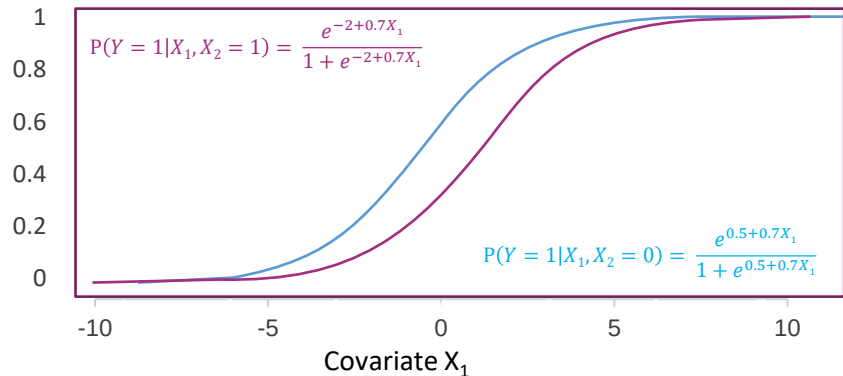
Multiple logistic regression

- Consider a multiple logistic regression model :

$$\log \left(\frac{p(x)}{1 - p(x)} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

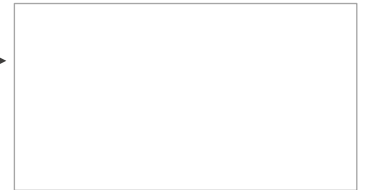
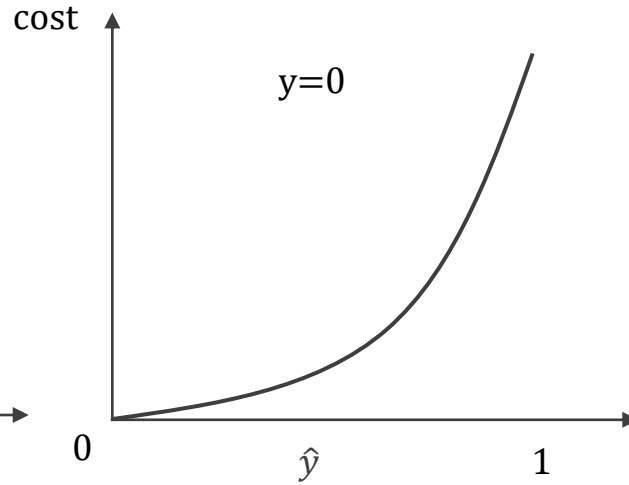
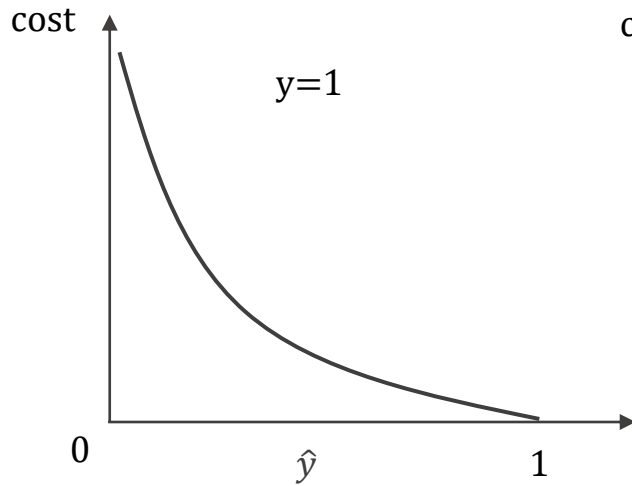
- Let X_1 be a continuous variable, X_2 a one hot encoded categorical variable taking binary value 0 or 1 .
- If the estimated values of parameters are:

$$\beta_0 = 0.5, \beta_1 = 0.7, \beta_2 = -2.5.$$



Loss/Cost Function Motivation

$$\text{Cost}(\hat{y}, y) = \begin{cases} -\log(\hat{y}), & \text{if } y = 1 \\ -\log(1 - \hat{y}), & \text{if } y = 0 \end{cases}$$



Loss Function (Binary Cross Entropy)

Another advantage of this loss function is that although we are looking at it by $y = 1$ and $y = 0$ separately, it can be written as one single formula which brings convenience for calculation:

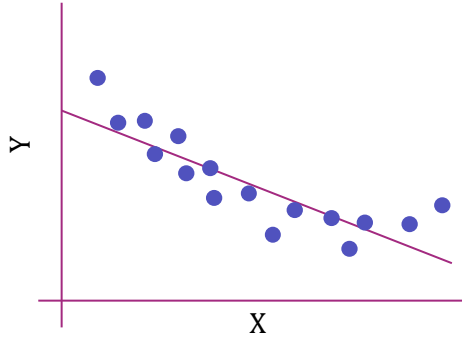
$$Cost(\hat{y}, y) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

So the cost function of the model is the summation from all training data samples:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m Cost(\hat{y}^{(i)}, y^{(i)})$$

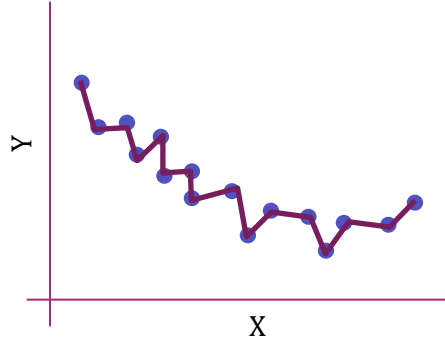
m=number of samples

Underfit



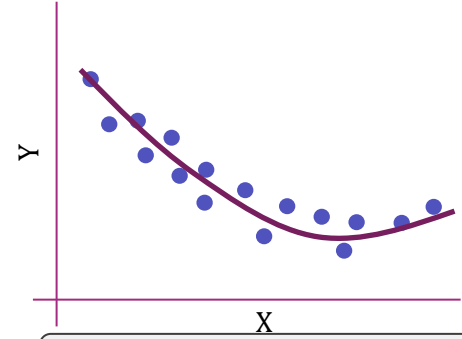
$$Y = \theta_0 + \theta_1 * X$$

Overfit

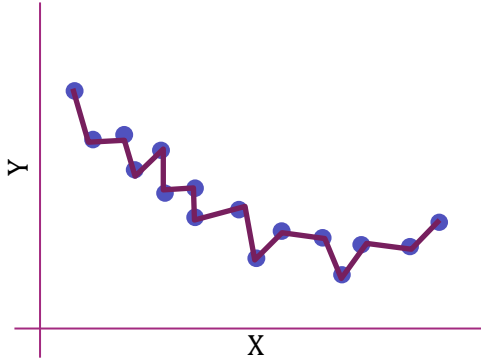


$$Y = \theta_0 + \theta_1 * X + \theta_2 * X^2 + \theta_3 * X^3 + \theta_4 * X^4$$

Balanced fit



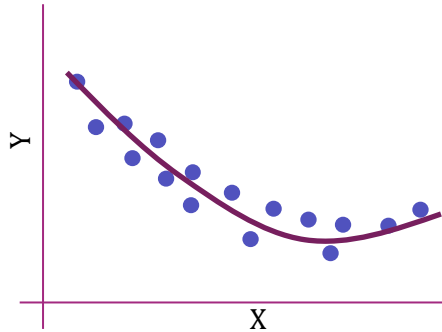
$$Y = \theta_0 + \theta_1 * X + \theta_2 * X^2$$



$$Y = \theta_0 + \theta_1 * X + \theta_2 * X^2 + \theta_3 * X^3 + \theta_4 * X^4$$



Try to make θ_3 and θ_4 almost close to zero



$$Y = \theta_0 + \theta_1 * X + \theta_2 * X^2$$

Suppose we penalize and make θ_1, θ_2 small

$$\min_{\theta} \frac{1}{2m} \sum_{i=1}^m \text{Cost}(\hat{y}^{(i)}, y^{(i)}) + 1000 \theta_1^2 + 1000 \theta_2^2$$

L2 Regularization Formula

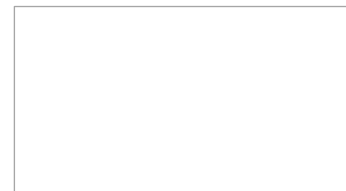
$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m \text{Cost}(\hat{y}^{(i)}, y^{(i)}) + \lambda \sum_{j=1}^n \theta_j^2$$

- Whenever we try to increase the value of theta, the error term increases so we add a penalty term which forces the theta to approach towards zero
- Lambda is the regularization parameter whose value is optimized for better results

$$\min_{\theta} \frac{1}{2m} \sum_{i=1}^m \text{Cost}(\hat{y}^{(i)}, y^{(i)}) + 1000 |\theta_1| + 1000 |\theta_2|$$

$$\text{Loss} = \frac{1}{n} \sum_{i=1}^n \text{Cost}(\hat{y}^{(i)}, y^{(i)}) + \lambda \sum_{i=1}^n |\theta_i|$$

- In L1 Regularization, we penalize the absolute value of the weights.
- Unlike L2 Regularization, the weights may be reduced to zero here.
- Hence, it is very useful when we are trying to compress our model/performing feature selection.



Standard Scaling

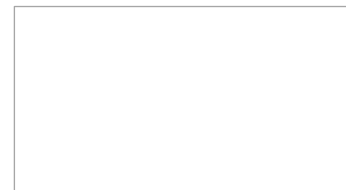
Standardization is another scaling technique where the values are centered around the mean with a unit standard deviation by computing Z-score. This means that the mean of the attribute becomes zero and the resultant distribution has a unit standard deviation.

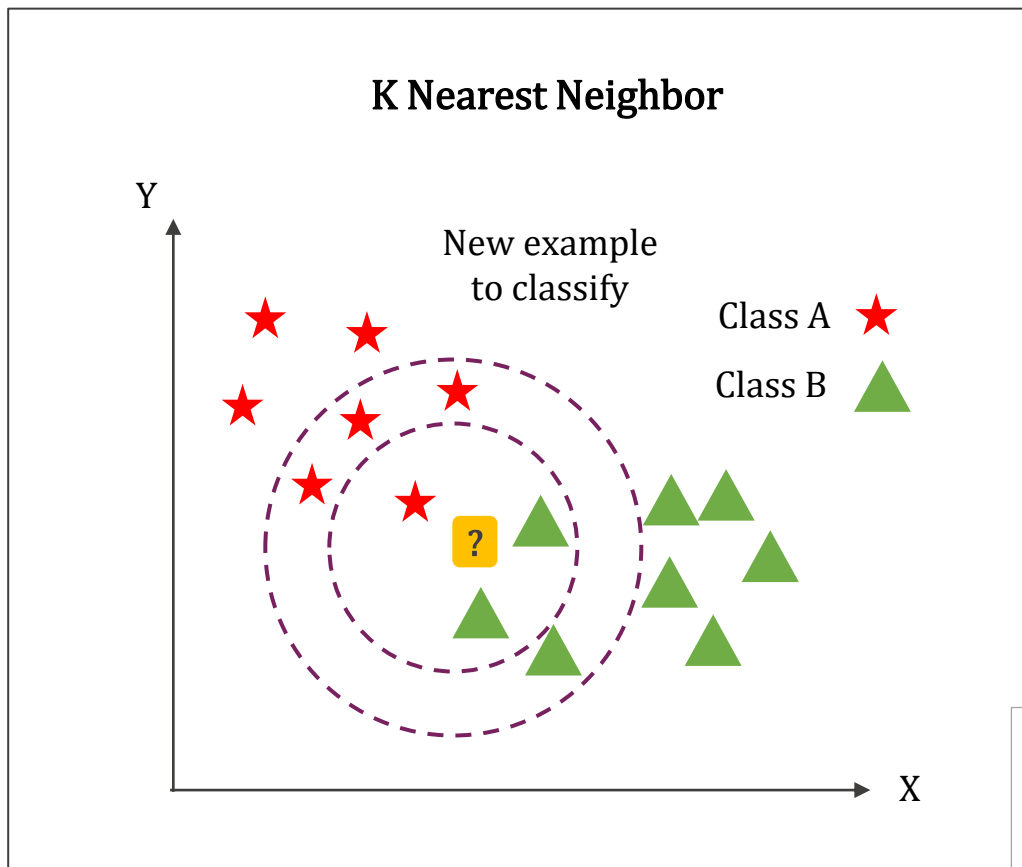
$$x_s = \frac{x - \mu}{\sigma}$$

Feature Normalization or Min-Max Scaling

Normalization is a scaling technique in which values are shifted and rescaled so that they end up ranging between 0 and 1. It is also known as Min-Max scaling.

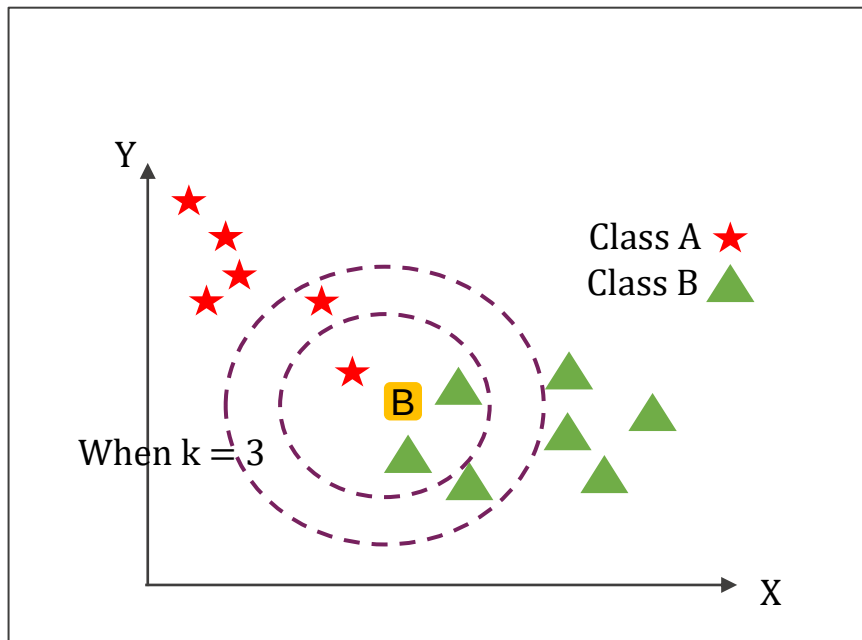
$$x_N = \frac{x - x_{min}}{x_{max} - x_{min}}$$





Core idea of K-Nearest Neighbors

- The KNN algorithm assumes that similar things exist in close proximity.
- KNN captures the idea of similarity by calculating the distance between closest points.



1. Identifying the cluster to which a particular Virtual Machine belongs depending on its CPU, Memory, I/O Bandwidth.

2. Recommender system

To find similar items, you compare the set of users who like each item—if a similar set of users like two different items, then the items themselves are probably similar!

3. Prediction of financial distressed companies based on early signs using K-nearest neighbours

1 Load the data

2 Initialize K to your chosen number of neighbors

3 For each datapoint in the dataset

- Calculate the distance between the query point and each datapoint.
- Maintain the calculated distance and index of the datapoint to an ordered collection

4 Sort the ordered collection of distances and indices from smallest to largest (in ascending order) by the distances

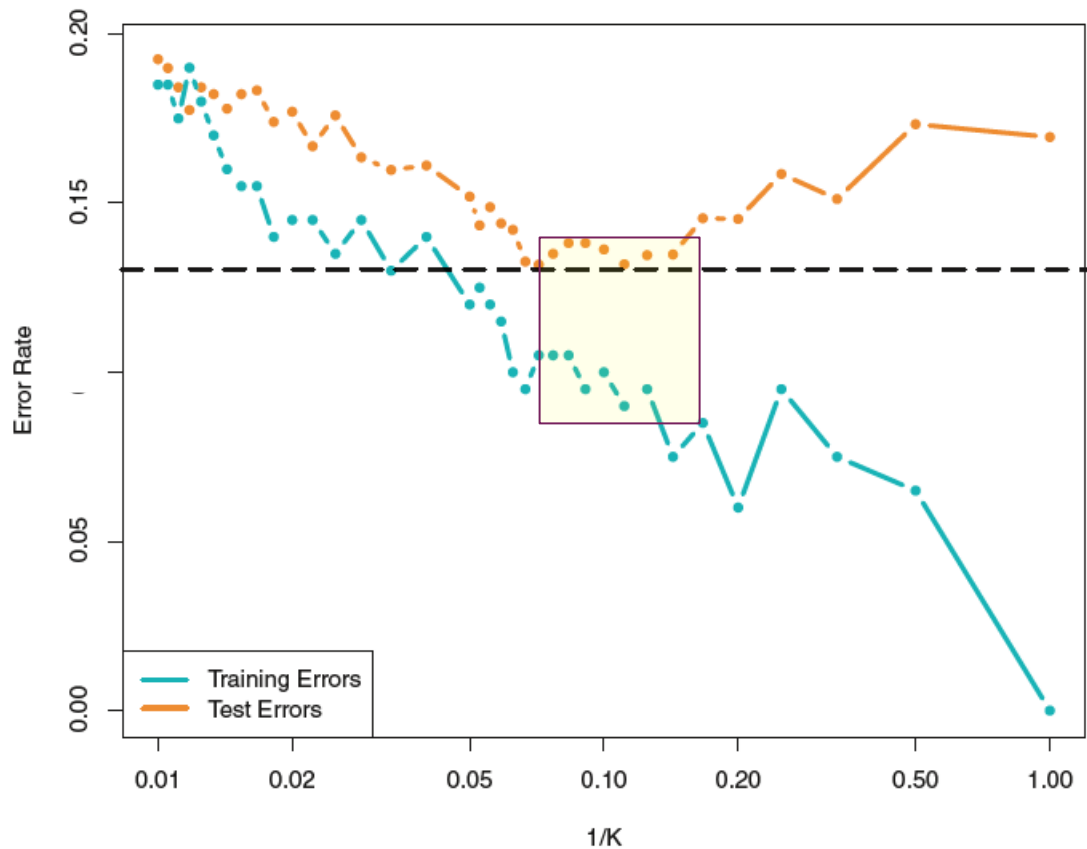
5 Pick the first K entries from the sorted collection

6 Get the labels of the selected K entries

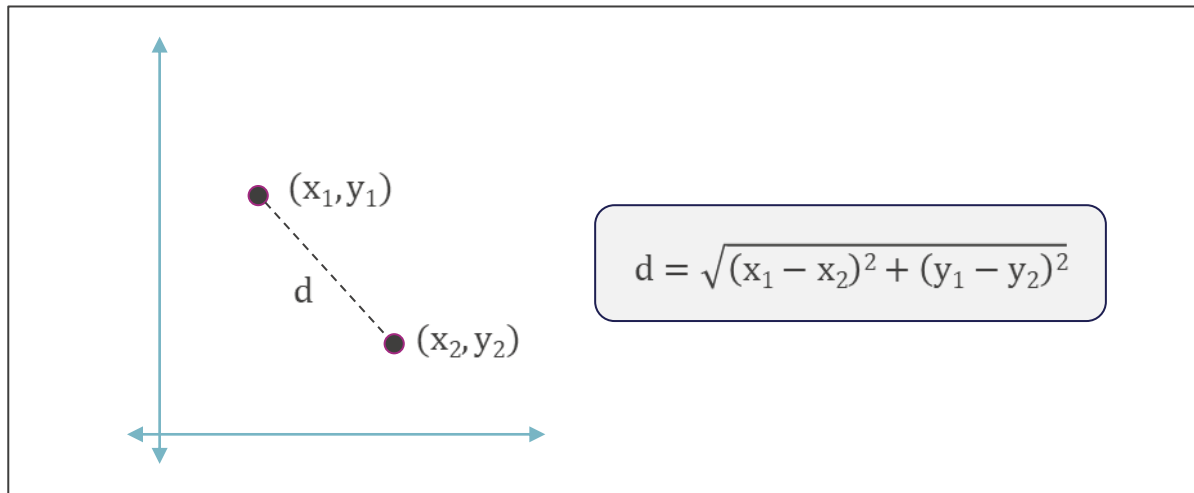
7 If regression, return the mean of the K labels

8 If classification, return the mode of the K labels

KNN – K Selection

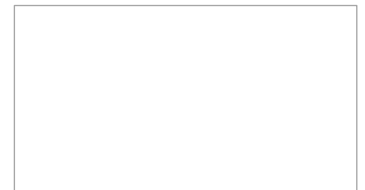


1. Euclidean distance:



2. Manhattan Distance:

$$M_{\text{dist}} = |x_2 - x_1| + |y_2 - y_1|$$



3. Hamming Distance

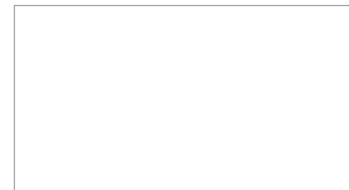
* Hamming distance is the count between where binary vectors differ.

Let x_1 and x_2 be two Boolean vectors

$$x_1 = [0, 1, 1, 0, 1, 1, 0]$$

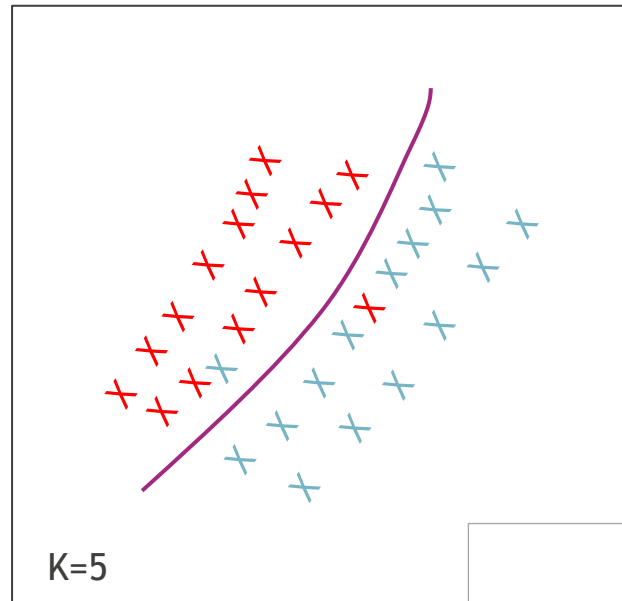
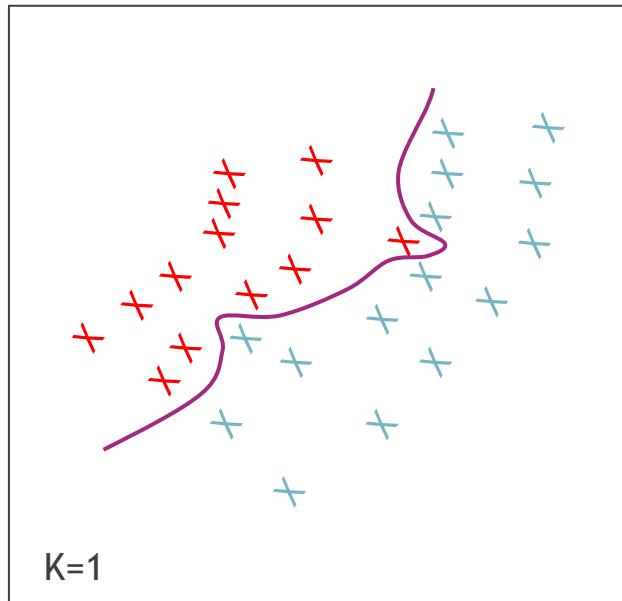
$$x_2 = [1, 1, 1, 0, 0, 1, 1]$$

$$\text{Hamming Distance} = 3$$



What Happens if K in KNN changes?

Drawing decision boundary



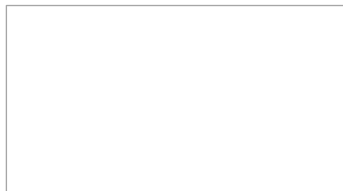
In cases where we are taking a majority vote (e.g. picking the mode in a classification problem) among labels, we usually make K an odd number to have a tiebreaker.

Advantages

1. The algorithm is simple and easy to implement.
2. There's no need to build a model, tune several parameters, or make additional assumptions.
3. The algorithm is versatile. It can be used for classification in different use cases

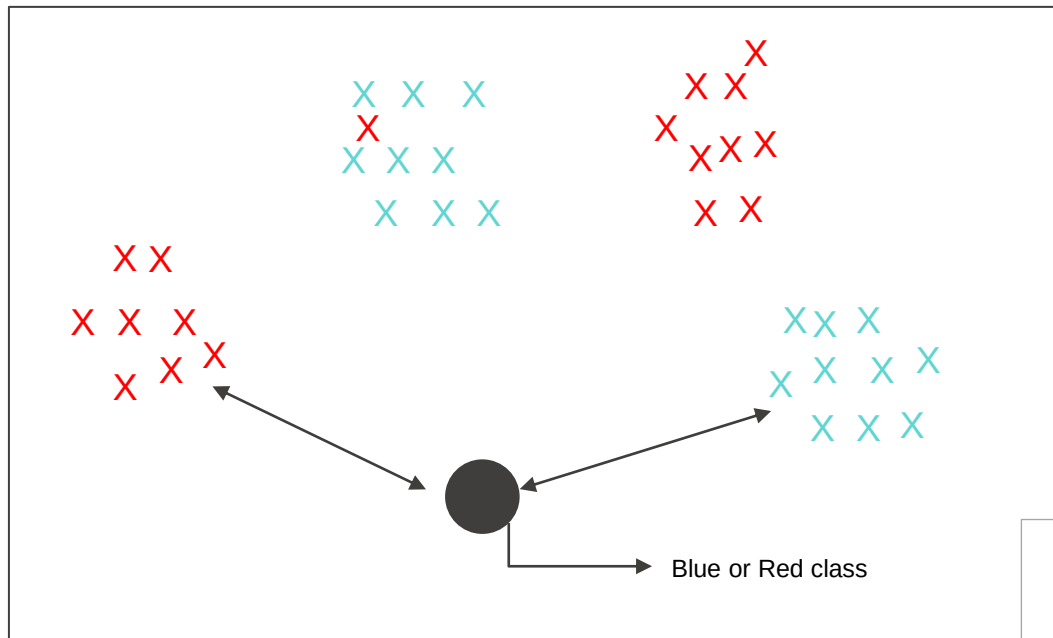
Disadvantages

1. The algorithm gets significantly slower as the number of examples and/or predictors/independent variables increase.



Failure Cases of KNN

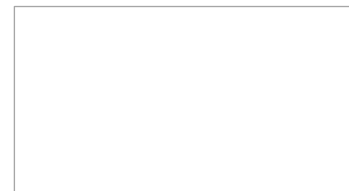
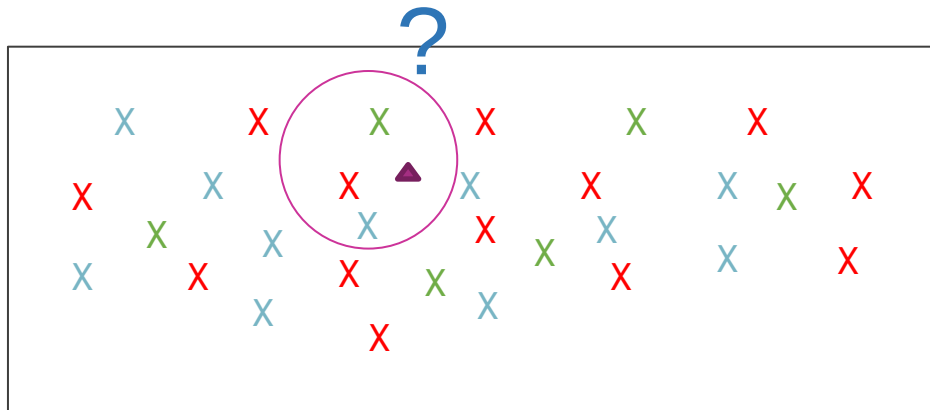
1. When the query point is far away from datapoints and somewhere in the middle of cluster of data points.



Failure Cases of KNN

Let's assume $k = 3$

2. If the points are randomly spread or jumbled up then there is no useful information at all and using KNN will not help.





Mathematics & Physics:

- Trigonometry: $\sin \theta = \frac{\text{OPP}}{\text{HYP}}$, $\cos \theta = \frac{\text{ADJ}}{\text{HYP}}$, $\tan \theta = \frac{\text{OPP}}{\text{ADJ}}$. Unit circle diagrams showing $\sin \theta$, $\cos \theta$, $\tan \theta$, $\csc \theta$, $\sec \theta$, $\cot \theta$.
- Calculus: $\frac{d}{dt} \int \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$, $\frac{d\phi_B}{dt} = h_f dx \frac{dB}{dt}$, $\frac{dE}{dx} = -h_f \frac{dB}{dt}$.
- Electromagnetism: $\vec{E} \cdot d\vec{l} = -d\phi_B$, $\vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$.
- Optics: Ray diagrams for lenses, $\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$, $m = \frac{s'}{s} = \frac{h'}{h}$.
- Wave Mechanics: $y = A \sin(\omega t + kx)$, $\frac{\partial^2 y}{\partial x^2} = -\frac{\partial^2 y}{\partial t^2}$.

Engineering & Electronics:

- Circuit diagrams: PN junction, transistor amplifier (Q1, Q2), AC source, OPEN CIRCUIT.
- Diagrams: $I_E = I_B + I_C$, $I_C = \alpha I_E$, $I_B = I_E - I_C$.

Chemistry & Biology:

- Photosynthesis: $6\text{CO}_2 + 12\text{H}_2\text{O} \xrightarrow{\text{light Chlorophyll}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.
- Protein Synthesis: Serine, Glycine, 2PGlycolate, 3PGlycolate, 1-6 DIAMINO HEXANOIC ACID, 1-6 DIHEXANOIC ACID.
- Chemical Structures: $\text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H}$ (PROPRANOIC ACID), $\text{H}-\text{C}-\text{C}-\text{O}-\text{H}$ (ETHANOL), $\text{H}-\text{C}-\text{C}-\text{O}-\text{C}-\text{C}-\text{H}$ (ETHYL PROPANOATE).

Set Theory & Logic:

- Venn diagrams: $A \cap B$, $A \cup B$, $A - B$, $A' \cap B$.
- Formulas: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Thank you

Credits: Trevor Hastie, Robert Tibshirani